GS FOUNDATION 2024-25

GEOGRAPHY - 29

Resources of India - II

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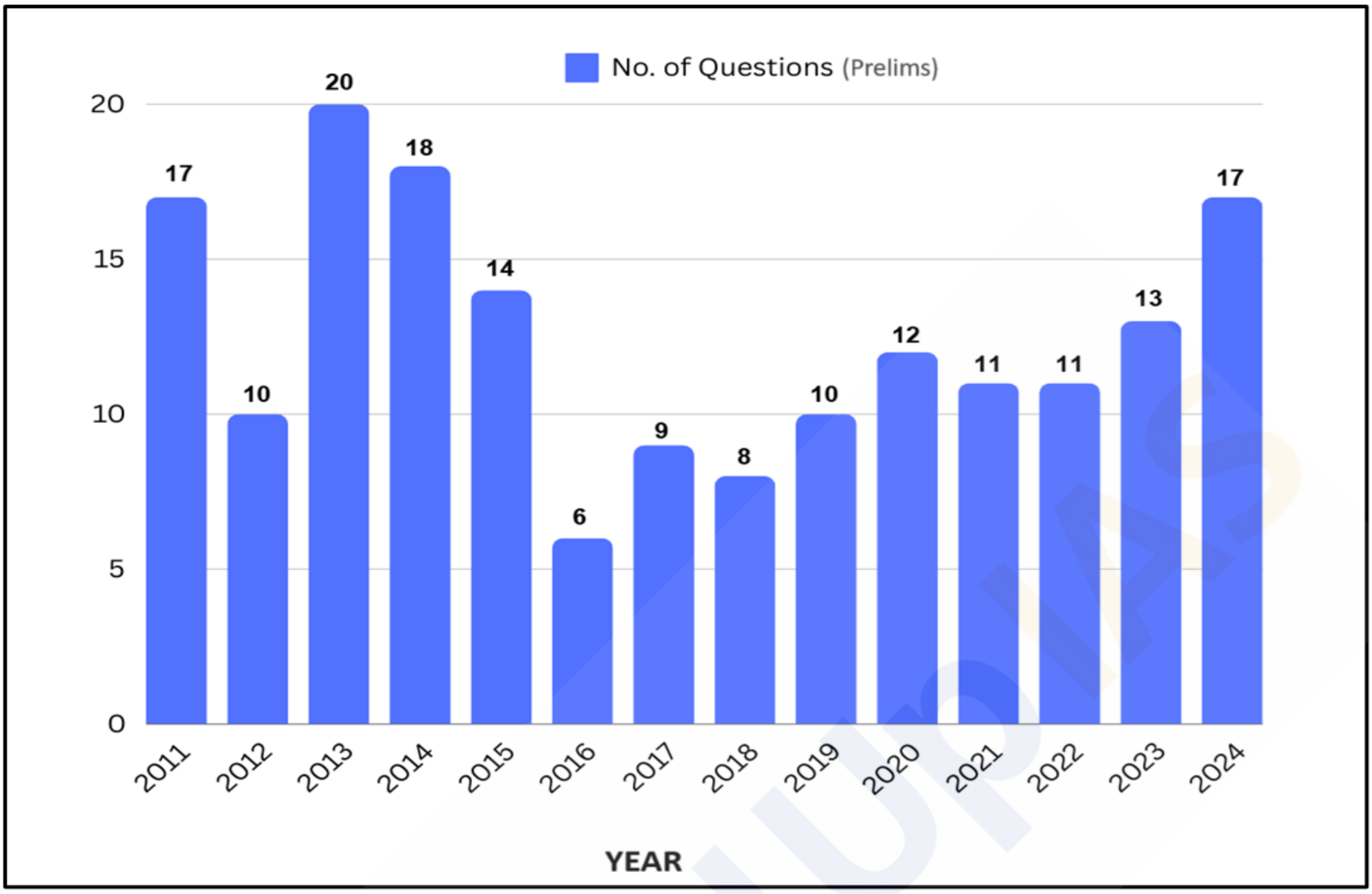

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Energy Resources

Introduction:

Energy is the **primary input** in the **production of goods and services,** serving as the driving force behind progress and development. A critical element in improving the **standard of living** for a country's population is the availability of **affordable and reliable energy services** in sufficient quantities. The more **consistent and abundant** the energy supply, the smoother the path toward **economic prosperity**.

The significance of energy has markedly increased with the rise of **industrialization** and **urbanization** in contemporary society. Initially, energy was primarily confined to household use as a cooking fuel. However, today it plays a crucial role across various sectors, including **industry**, **commerce**, **transport**, and **telecommunications**. Furthermore, it is essential for a wide array of **household services**, reflecting its extensive impact on daily life and economic activity.

Classification of Energy Sources:

Energy can be categorized into two broad classes based on its source and utilization:

- **Traditional or Non-Commercial Energy -** This category includes sources such as **firewood**, **charcoal**, **cow dung**, **agricultural waste**, and **animal power**. These energy forms are typically used in rural areas and are often less efficient and less reliable.
- **Commercial Energy -** This includes more modern and widely utilized sources such as **coal**, **oil**, **natural gas**, **hydroelectricity**, **nuclear power**, as well as **wind** and **solar power**. These sources are integral to the functioning of contemporary economies.

Additionally, energy can also be classified into conventional and non-conventional types based on its nature:

- **Conventional Energy -** These are the **traditional sources that have been used for centuries.** They are **generally non-renewable,** meaning they **take millions of years** to form and cannot be replenished at a rate that keeps up with consumption. This category encompasses sources such as **coal**, **petroleum**, **natural gas**, and **electricity**. These sources have been the backbone of energy consumption for many decades.
- **Non-Conventional Energy -** These are also known as **alternative or renewable energy sources.** They are generally **cleaner and more sustainable** than conventional sources. This includes **alternative sources** like **solar**, **wind**, **tidal**, **geothermal energy**, and **biogas**. These emerging forms of energy are becoming increasingly important due to their potential to reduce reliance on fossil fuels and mitigate environmental impact.

Conventional Sources of Energy:

As previously mentioned, the primary **conventional sources of energy** include **coal**, **petroleum**, **natural gas**, and **electricity**. Each of these sources has its own production, distribution, and consumption characteristics, which will be explored in further detail in the following sections. Understanding these sources is vital for appreciating their roles in the broader context of energy availability and economic growth.

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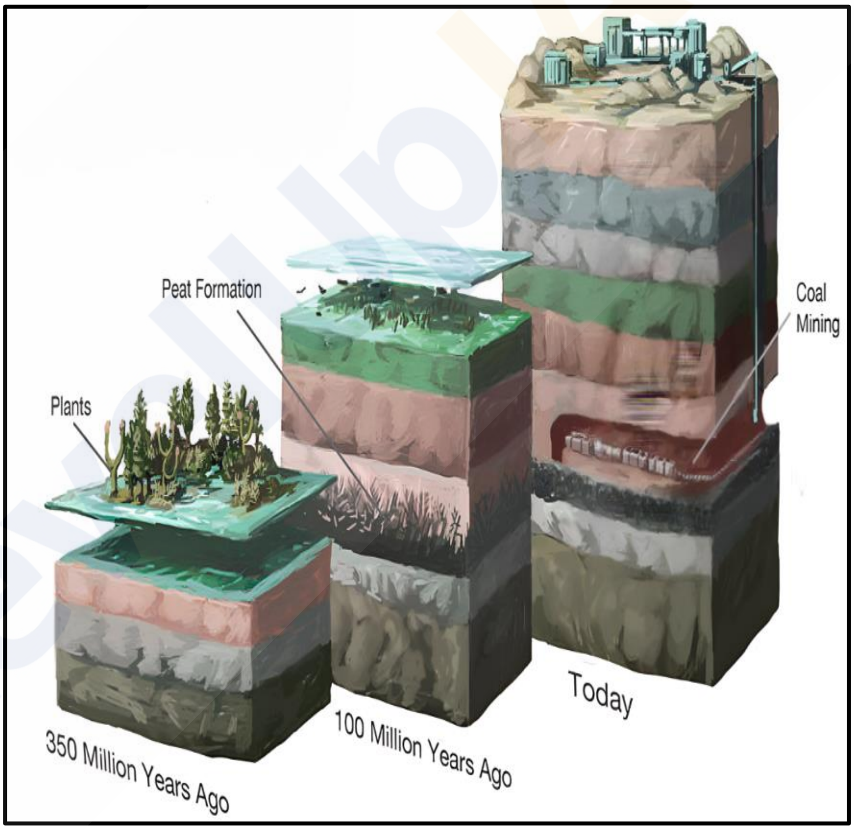
Coal:

- Coal is a **combustible organic substance** primarily composed of **hydrocarbons.** It exists in the form of **sedimentary rocks** and serves as a vital source of fuel for generating heat, light, or both. In addition to hydrocarbons, coal contains **volatile matter**, **moisture**, and **ash** in varying proportions. The **combustible components** of coal are primarily **carbon** and **hydrogen**.
- **In India,** coal has been, is, and will continue to be the **mainstay of power generation** for an extended period. It constitutes approximately **70%** of the total **commercial energy** consumed in the country. A significant portion of coal consumption—**94%** —is attributed to the **power sector and industries.** Essential industries such as **iron and steel manufacturing** and various **chemical productions** heavily depend on coal availability. Due to its critical role as an energy source and raw material for numerous industries, coal is often referred to as **black gold**. Recent studies by energy experts indicate that global coal reserves are **six times greater** than the known reserves of oil, underscoring coal's position as a **bridge into the future**.

Origin of Coal:



- The formation of coal is linked to **organic matter**, particularly wood. During the **Carboniferous age**, vast tracts of forest lands were buried under sediments. The **burning and decomposition** of wood, caused by heat from below and pressure from above, initiated the transformation process.
- As wood decomposed, **hydrogen** was released in the form of **methane** and **water**, while **oxygen** was released as **carbon dioxide** and **water**. Throughout this transition from wood to coal, the **amount of oxygen and nitrogen decreased**, while the **proportion of carbon increased**. The energy content of coal is directly related to its **carbon percentage**, which is influenced by the **duration and intensity of heat and pressure** applied to the wood.

Varieties of Coal:

Anthracite Coal:

- Anthracite is regarded as the **best quality of coal,** containing **80 to 95% carbon**. It has minimal volatile matter and a negligible proportion of moisture. This type of coal is **hard, compact,** and has a **jet-black** appearance with a **semi-metallic lustre.** Anthracite ignites slowly and burns with a **short blue flame,** providing the **highest heating value** among all coal varieties. In India, anthracite is found **only in Jammu and Kashmir,** and in limited quantities.

Bituminous Coal:

- Bituminous coal is the **most widely used** type of coal and derives its name from **bitumen**, a liquid released during heating. Its carbon content ranges from **40 to 80%**, with **moisture and volatile content** varying between **15 to 40%**. It is often subdivided into **sub-bituminous** and **bituminous coals**. Bituminous coal is dense, compact, and typically **black in colour,** displaying alternating dull and bright bands. It lacks traces of the original vegetable material from which it was formed. Due to its high calorific value, bituminous coal is utilized for **steam raising**, **heating**, and the production of **coke** and **gas**. Major reserves of bituminous coal are found in **Jharkhand**, **Odisha**, **West Bengal**, **Chhattisgarh**, and **Madhya Pradesh**.

Lignite:

- Also known as **brown coal**, lignite is a **lower-grade coal** containing **40 to 55% carbon**. It represents an **intermediate stage** in the transformation of woody matter into coal. Lignite's colour ranges from **dark** to **black brown** and is **friable** and **pyritic**. It has a **high moisture content** (over **35%**), resulting in significant smoke production with limited heat output. Lignite is prone to disintegration when exposed to air and can even undergo **spontaneous combustion**. Major deposits are found in **Palna** (Rajasthan), **Neyveli** (Tamil Nadu), **Lakhimpur** (Assam), and **Karewa** (Jammu and Kashmir).

Peat:

- Peat represents the **initial stage** of wood transformation into coal and contains less than **40 to 55% carbon**, along with substantial volatile matter and moisture. Peat is rarely compact enough to serve as a good fuel source without being compressed into **bricks**. When left in its natural state, peat burns like wood, producing less heat, emitting more smoke, and leaving a significant amount of ash after combustion.

Occurrence of Coal in India:

- **India's coal-bearing strata** can be categorized geologically into **two primary groups - Gondwana Coal Fields** and **Tertiary Coal Fields.**

Gondwana Coal Fields:

- The Gondwana coal fields constitute an overwhelming **majority of India's coal reserves and production,** accounting for **98% of total reserves** and **99% of total production**. These fields are recognized as a significant repository of both **metallurgical and high-quality coal**. Out of the **113 major coal fields** in India, **80 are situated within the rock systems of the lower Gondwana Age**. Approximately **75 separate basins** span an area of **77,700 square kilometers**, predominantly located in **Peninsular India**. The size of these basins varies significantly, ranging from **1 square kilometer** to **1,550 square kilometers**.

- The Gondwana coal basins are primarily located in the valleys of several rivers, including the **Damodar** (Jharkhand-West Bengal), **Mahanadi** (Chhattisgarh-Odisha), **Son** (Madhya Pradesh-Jharkhand), **Godavari**, **Wardha** (Maharashtra-Andhra Pradesh), **Indravati**, **Narmada**, **Koel**, **Panch**, **Kanhan**, and others.
- Dating back about **250 million years**, Gondwana coal comprises both **coking and non-coking varieties**, as well as **bituminous and sub-bituminous coal**. However, **anthracite** coal is generally absent from these fields. The coal contains **volatile compounds and ash** that typically range from **13% to 20%**, sometimes increasing to **25% to 30%**, which limits the carbon content to a maximum of **55% to 60%**. Furthermore, Gondwana coal is almost moisture-free, although it does contain small, variable amounts of **sulphur and phosphorus**.
- It is also noteworthy that some coal-bearing Gondwana formations may be concealed beneath the extensive lava flows of the **Deccan Trap**. In regions like the **Satpuras**, erosion has exposed the Gondwana strata, indicating the potential for substantial quantities of valuable coal extraction in these areas. The **Damuda series** (Lower Gondwana) represents the most significant and extensively developed coalfields, contributing to **approximately 80% of India's total coal production**.

Tertiary Coal Fields:

- In contrast, the Tertiary coal fields are **geologically younger,** with ages ranging from **15 to 60 million years**. These coal deposits are primarily found in areas outside the Peninsular region. Tertiary coal generally exhibits a **lower carbon content** and a **higher percentage of moisture and sulphur** compared to Gondwana coal. Key regions rich in Tertiary coal include parts of **Assam**, **Meghalaya**, **Arunachal Pradesh**, **Nagaland**, the **Himalayan foothills** of **Darjeeling** (West Bengal), as well as areas in **Jammu and Kashmir**, **Uttar Pradesh**, **Rajasthan**, **Kerala**, **Tamil Nadu**, and the union territory of **Puducherry.**

Production and Distribution of Coal in India:

Gondwana Coalfields:

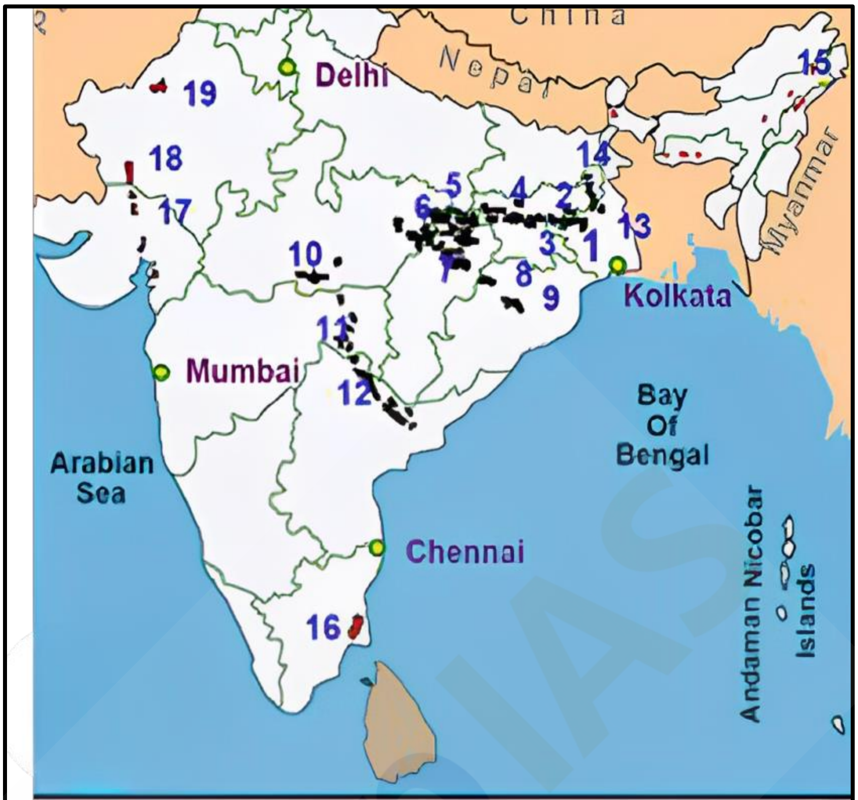
- The Gondwana coal, being older, is more significant in terms of reserves and production. The state-wise distribution of major Gondwana coal-producing areas is described below:

Jharkhand:

- Jharkhand occupies the **top spot with a staggering 26.4% share of India's total coal reserves. It is the third largest producer of coal in India.** It is one of the most important coal-producing states in India, with the majority of its coalfields located in a narrow east-west belt along the 24°N latitude.
    - **Jharia Coalfield -** Located southwest of **Dhanbad city,** this coalfield covers an area of **453 sq km** and is one of the oldest and richest coalfields in India. It is renowned as the **storehouse of the best metallurgical coal** in the country.
    - **Bokaro Coalfield -** Situated in **Hazaribagh district,** it lies just 32 km west of the Jharia coalfield. Bokaro is another crucial coal-producing region in Jharkhand.
    - **Other Reserves -** Additional coal reserves in Jharkhand include **Giridih** and **Karanpura** coalfields, which contribute to the state's significant coal output.

Odisha:

- Odisha, with its abundant coal reserves, ranks as the **second-largest state by reserves** and the **largest coal producer,** contributing around **25%** of India's total coal production.
    - **Talcher Coalfield** - Stretching from **Talcher town to Rairkhol** in **Dhenkanal and Sambalpur districts,** this coalfield is second only to **Raniganj** in terms of reserves. Most of the coal mined here is used in thermal power and fertilizer plants at Talcher.
    - **Other Coalfields** - Notable coalfields in Odisha include **Rampur-Himgir** in Sambalpur district and the **Ib River Coalfield** in the districts of Sambalpur and Gangpur.



GONDWANA COALFIELDS			
1.	Raniganj	2.	Jharia
3.	Bokaro	4.	North Karanpura
5.	Singrauli	6.	Sohagpur
7.	Korba	8.	Ib-valley
9.	Talchir	10.	Satpura
11.	Wardha	12.	Godavari
13.	Birbhum	14.	Rajmahal
TERTIARY COAL / LIGNITE FIELDS			
15.	Assam-Mehgalaya	16.	Neyveli
17.	Cambay	18.	Barmer-Sanchor
19.	Bikaner	20.	Jammu and Kashmir

Chhattisgarh:

- Chhattisgarh holds the **third position** in terms of **coal reserves** but ranks **second in production,** after Odisha. The state's coalfields are crucial for India's energy supply.
    - **Korba Coalfield -** Spread over **515 sq km**, this coalfield is located in the **Hasdo River valley** (a tributary of the Mahanadi) in Korba district. It is one of the most important coalfields in Chhattisgarh.
    - **Other Coalfields** - Key coalfields in Chhattisgarh include **Hasdo-Arand**, **Chirmiri**, **Jhilmili**, and **Johila**, contributing to the state's overall coal production.

Madhya Pradesh:

- Madhya Pradesh is the **fourth-largest** coal-producing state in India, with significant deposits in its central regions.
    - **Singrauli Coalfield -** Situated in **Sidhi** and **Shahdol** districts, this coalfield is the largest in the state and supplies coal to major thermal power plants in **Singrauli** and **Obra**.
    - **Pench-Kanhan-Tawa Coalfields -** Located in **Chhindwara district**, this coalfield is another major contributor to Madhya Pradesh's coal output.

Andhra Pradesh and Telangana:

- Andhra Pradesh and Telangana together account for **9.72%** of India's coal production, with most of the reserves located in the **Godavari Valley**.
    - **Singareni and Kothagudem Collieries -** These are the most significant coal-producing areas in the region, situated in **Adilabad**, **Karimnagar**, **Warangal**, **Khammam**, and **East and West Godavari districts**. The coal here is predominantly of the **non-coking variety**, making it a crucial source of coal for southern India.

Maharashtra:

- Maharashtra, despite having only **3%** of India's coal reserves, contributes over **9%** to the nation's total coal production.
    - **Kamptee Coalfields -** Located in **Nagpur district**, these coalfields, along with the coal reserves of **Wardha Valley**, **Ghughus**, **Ballarpur**, and **Warora** in **Chandrapur district**, are important coal-producing regions.
    - **Wun Field -** Situated in **Yavatmal district**, the Wun coalfield is another significant coal-producing area in Maharashtra.

West Bengal:

- Although West Bengal produces only **6%** of India's coal, the state holds over **11%** of the country's coal reserves, mainly concentrated in the **Raniganj Coalfield**.
    - **Raniganj Coalfield -** Located in **Burdwan district**, Raniganj is the **largest coalfield in West Bengal** and is known for its extensive reserves.
    - **Other Coalfields -** Besides Raniganj, coal is also mined in **Bankura**, **Purulia**, **Birbhum**, **Darjeeling**, and **Jalpaiguri** districts.

Tertiary Coalfields:

- Tertiary coal-fields in India are primarily associated with **limestone and slates** from either the **Eocene** or **Oligocene-Miocene** ages. These coal deposits are notable for their geological occurrence and distribution across **several northeastern states.**

Assam:

- In Assam, the major coalfields include **Makum** and **Nazira**, with the **Makum coalfield** in the **Sibsagar district** being the most developed and significant. Assam's coal is characterized by its **low ash content** and **high coking quality**. However, due to its **high Sulphur content**, it is not suitable for metallurgical purposes.

Meghalaya:

- In Meghalaya, deposits of tertiary coal are found in the **Garo**, **Khasi**, and **Jaintia hills**, which are believed to belong to the **Lower Eocene** age. These deposits are important for the state's coal production.

Arunachal Pradesh:

- The **Upper Assam Coal belt** extends into Arunachal Pradesh as the **Namchick-Namrup coalfield** in the **Tirap district.** This region marks the eastern extension of Assam's coal resources.

Other Regions:

- In addition to the northeastern states, tertiary coalfields are also found in **Jammu and Kashmir** and **Himachal Pradesh**, though their economic significance is less compared to the northeastern coalfields.

Lignite Coalfields:

- There has been a significant increase in **lignite production** in India over the years. Although lignite deposits are distributed across several states, **Tamil Nadu** dominates in terms of both **reserves** and **production**.

Tamil Nadu:

- Tamil Nadu accounts for **90 percent of India's lignite reserves** and approximately **71 percent of its production**. The most important lignite field is the **Neyveli Lignite fields** in the Cuddalore district, which cover an area of 480 sq km and have estimated reserves of **4,150 million tonnes**. Other notable lignite reserves in Tamil Nadu include **Jayamkondacholapuram** in the Trichy district, **Mannargudi**, and areas **east of Veeranam**.

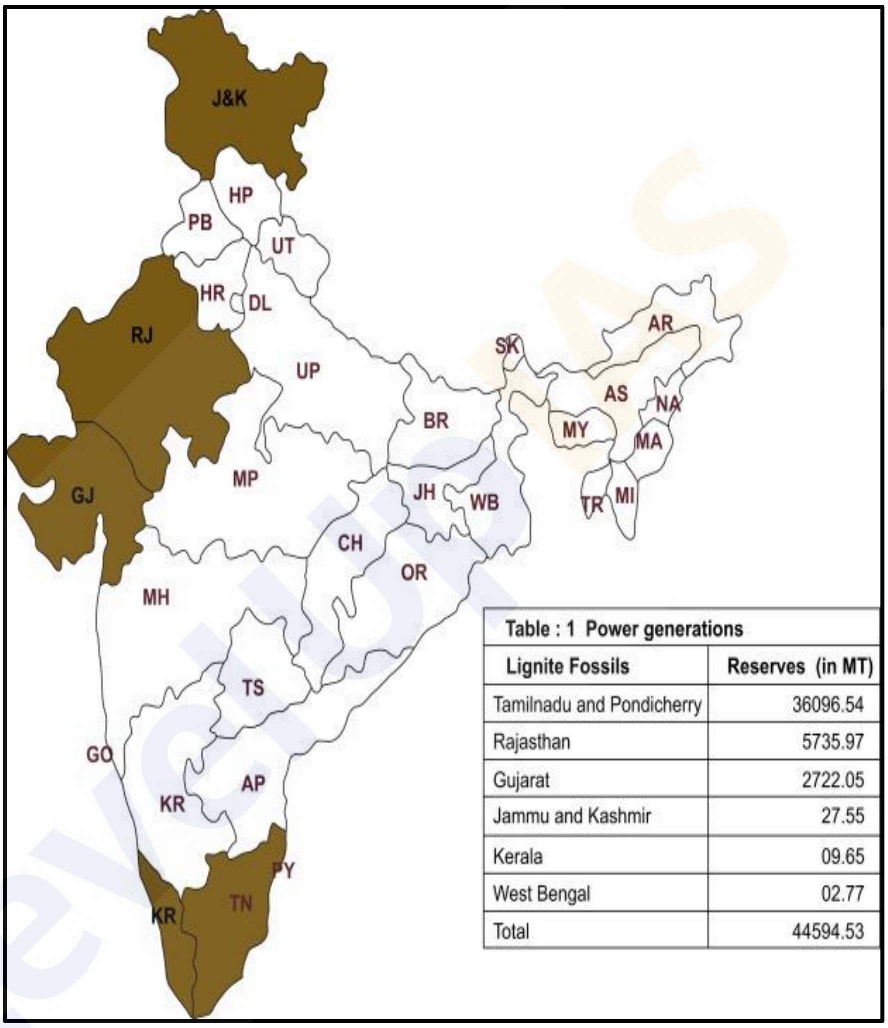


Table : 1 Power generations	
Lignite Fossils	Reserves (in MT)
Tamilnadu and Pondicherry	36096.54
Rajasthan	5735.97
Gujarat	2722.05
Jammu and Kashmir	27.55
Kerala	09.65
West Bengal	02.77
Total	44594.53

Gujarat:

- In Gujarat, lignite deposits are found in the **Kachchh district** at locations such as **Umarsar**, **Lefsi**, **Jhalrai**, and **Baranda**, as well as in the **Bharuch district**.

Jammu and Kashmir:

- Jammu and Kashmir have lignite deposits dating back to the **Pliocene** or even more recent periods. The main lignite fields are found in the **Shaliganga River** area, extending northwest to the **Nichahom area** in the Handwara region of the Baramula district. However, the lignite found here is of **poor quality**.

Other States:

- Lignite is also produced in **Kerala**, **Rajasthan**, **West Bengal**, and **Puducherry**, though the quantities are relatively small compared to Tamil Nadu and Gujarat.

Peat Coalfields:

- **Peat** deposits are confined to a **few specific areas in India.** These include:
  - **Nilgiri Hills -** Peat occurs at an elevation of over 1,800 meters.
  - **Kashmir Valley -** Peat is found in the alluvium of the **Jhelum River** and swampy grounds in the higher valleys.
  - **West Bengal -** In Kolkata and its suburbs, peat beds have been found at depths ranging from 2 to 11 meters.
  - **Ganga Delta -** The delta region contains layers of peat, primarily composed of **forest** and **rice plants**.

Problems of Coal Mining in India:

Uneven Distribution of Coal Reserves:

- The geographical distribution of coal in India is highly uneven. **Major coal-producing areas** are concentrated in the states of **Jharkhand, Chhattisgarh, Madhya Pradesh, Odisha, and West Bengal**. In contrast, most of the northern plains and western regions of India have little or no coal reserves. This **geographic disparity** leads to **high transportation costs** as coal, being a heavy commodity, must be transported over long distances to reach industries located far from the mining areas. As a result, **coal-consuming industries in distant regions face higher prices**, impacting their competitiveness and overall cost of production.

Low Calorific Value and High Ash Content:

- Indian coal is characterized by **low calorific value** and **high ash content**, with the ash content ranging from **20% to 30%**, and in some cases, even exceeding **40%**. This significantly reduces the **energy output of coal**, making it less efficient as a fuel source. Additionally, the high ash content creates complications in **ash disposal**, posing environmental challenges. **Low-quality coal** thus increases both operational costs and environmental concerns for industries relying on coal as a primary energy source.

Low Productivity in Underground Mines:

- A significant proportion of India's coal is extracted from **underground mines**, where both **labour** and **machinery productivity** are notably low. Despite large investments aimed at modernizing these mines, the **output per man shift (OMS)** has stagnated at around **0.55 tonnes** for the last two decades. These underground mines employ **80% of the workforce** but contribute only **30% of total coal production**. Consequently, the **cost of coal production** has increased dramatically, from **₹50 per tonne in 1973-74** to **₹550 per tonne in 2011-12**, making coal mining a less profitable venture and driving up costs for industries reliant on coal.

Losses Due to Fires and Pilferage:

- India's coal mining sector suffers from significant **losses due to fires in the mines** and at pit-heads. Fires can lead to the loss of large quantities of coal, while **pilferage at various stages** of coal production and transportation further exacerbates the problem. These factors contribute to a **rise in coal prices**, which in turn affects the wider economy by creating a **vicious cycle of price increases** across industries dependent on coal.

Environmental Pollution:

- Coal mining and utilization pose severe environmental challenges. **Open-cast mining** devastates the landscape, transforming the area into **rugged and ravinous land**, which is difficult to reclaim. Additionally, **coal dust** generated in mines and near pit-heads creates **serious health hazards** for mine workers and their families. The **burning of coal** in thermal power plants and industrial factories releases **toxic gases** into the atmosphere, contributing to air pollution. Implementing **safety and pollution control measures** to mitigate these environmental impacts is **costly and complex**, often beyond the reach of smaller or ordinary enterprises, making it difficult to balance industrial growth with environmental sustainability.

Conservation of Coal in India:

The **conservation of coal** has emerged as a critical concern in India, especially due to the challenges posed by the inefficient use of coal in various sectors. The **misuse of good quality coal** for burning in transportation and industries, coupled with the **short lifespan of metallurgical coal**, are key concerns. Additionally, **selective mining** practices have led to large-scale wastage of raw coal, while **frequent fires in mines** and **unscientific methods of coal extraction** further exacerbate the issue. Therefore, conservation strategies are imperative to ensure the efficient and sustainable use of coal resources in India.

Coal conservation refers to the **efficient utilization of every bit of energy that can be extracted from coal,** along with the **recovery of every possible by-product.** It is an integral aspect of **mine planning and operations to maximize the value obtained from coal resources while minimizing wastage.** Effective conservation not only helps in extending the life of coal reserves but also contributes to environmental sustainability by reducing the negative impacts of coal mining and use.

Suggested Measures for Coal Conservation:

- **Use of Coking Coal for Metallurgical Industry Only -** Coking coal, a high-quality variety of coal primarily used in the steel industry, should be reserved exclusively for **metallurgical purposes**. This ensures that the most valuable form of coal is not wasted in other industries that could function with lower-grade coal.
- **Washing and Blending of Low-Grade Coal -** Lower-grade coal, which contains impurities and offers less energy, should undergo a **washing process** to improve its quality. It should then be **blended with superior quality coal** in appropriate proportions before being used in industries, thus optimizing the use of all available coal resources.
- **Discouragement of Selective Mining Practices - Selective mining**, which involves extracting only the best quality coal while leaving lower-quality coal behind, should be discouraged. Such practices lead to significant wastage of coal that could otherwise be utilized. All possible coal from mines should be extracted, regardless of its grade, to minimize resource loss.
- **Discovery of New Reserves and Adoption of Modern Techniques -** Efforts must be made to **discover new coal reserves** through exploration and to adopt **advanced techniques** that enable more efficient extraction and use of coal. New technologies can enhance recovery rates and improve the sustainability of coal mining.
- **Amalgamation of Small and Uneconomic Collieries -** Small and uneconomic collieries, which operate at a loss or produce very low quantities of coal, should be **amalgamated** to form larger, more economically viable units. This consolidation can lead to better resource management, improved efficiency, and enhanced profitability.

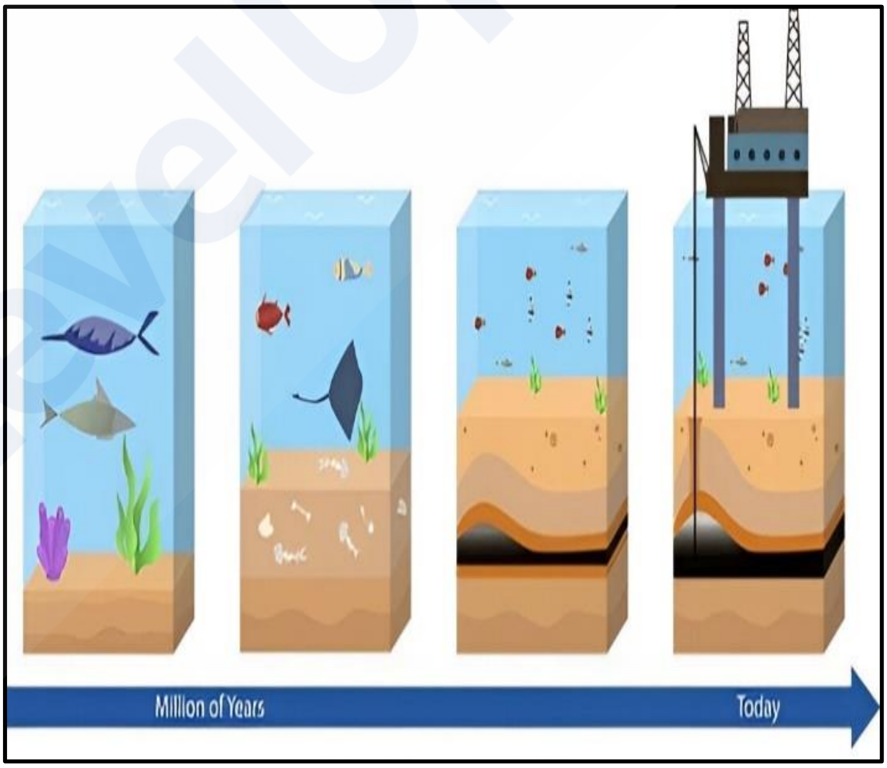
The conservation of coal in India is a multifaceted challenge that requires careful planning and strategic implementation of various measures. By prioritizing the **efficient use of coking coal**, **improving the quality of low-grade coal**, **discouraging selective mining**, and **exploring new reserves**, India can make significant strides in ensuring that its coal resources are used in a sustainable and efficient manner. Moreover, **modernization of mining techniques** and the **amalgamation of smaller collieries** will further contribute to the overall conservation effort.

Petroleum or Mineral Oil:

- The term **'petroleum'** is derived from two Latin words: ***Petra*** (meaning rock) and ***Oleum*** (meaning oil). Hence, petroleum refers to **oil obtained from rocks,** particularly from sedimentary rocks of the Earth. This is why petroleum is also called **'Mineral Oil.'** From a technical perspective, petroleum is an inflammable liquid predominantly composed of **hydrocarbons**, which make up **90 to 95%** of its composition. The remaining components include **organic compounds** containing **oxygen, nitrogen, sulfur**, and traces of **organo-metallic compounds**. **Crude petroleum** is a mixture of hydrocarbons in solid, liquid, and gaseous states. It includes compounds from the **paraffin series**, some unsaturated hydrocarbons, and a small proportion of compounds from the **benzene group**.
- Petroleum and its derivatives serve as one of the most crucial sources of **motive power**. It is a **compact and convenient liquid fuel** that has revolutionized transportation on land, air, and water. Its easy transportability from production to consumption areas, whether through **tankers** or more efficiently via **pipelines**, makes it a highly convenient fuel source. Additionally, petroleum emits minimal smoke and leaves no ash (unlike coal), ensuring it can be utilized efficiently down to the last drop. Moreover, petroleum is essential for providing **lubricating agents** and serves as a critical raw material for various **petrochemical products**.

Origin and Occurrence of Petroleum:

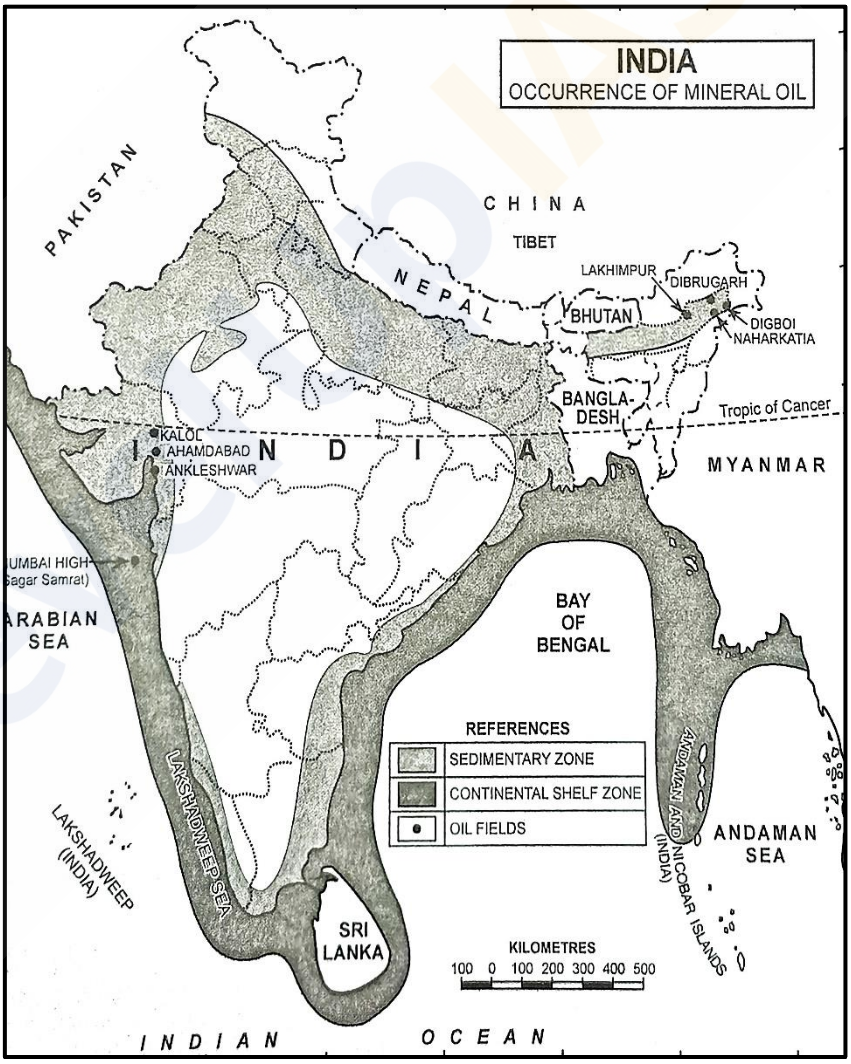
- Petroleum has an **organic origin** and is primarily found in **sedimentary basins, shallow depressions**, and sometimes **beneath the seas,** both past and present. In India, most oil reserves are associated with **anticlines** and **fault traps** within sedimentary rock formations that date back to **Tertiary times**, approximately **3 million years ago.** Some more recent sediments, less than one million years old, also show signs of **incipient oil** formation. Oil and **natural gas** are believed to have originated from the decomposition of **animal and vegetable matter** in shallow marine sediments, such as **sands, silts,** and **clays,** which were deposited during periods of abundant land and aquatic life. In particular, **microscopic flora and fauna** were integral to this process.



- **Conditions for oil formation** were especially favorable during the **Lower and Middle Tertiary period**, when dense forests and marine organisms thrived in **gulfs, estuaries, deltas**, and the surrounding landmasses. The **decomposition of organic matter** in these **sedimentary rocks** eventually led to the **formation of petroleum.** While oil is predominantly found in sedimentary rocks, not all sedimentary formations contain oil.

- **For an oil reservoir to form, three critical conditions must be met:**
    - **Porosity** to accommodate sufficient amounts of oil.
    - **Permeability** to allow oil or gas to flow out when drilled.
    - **Impervious beds** above the porous sandstone, conglomerates, or fissured limestone layers, to prevent oil from dissipating through percolation into surrounding rocks.
- **Commercially viable oil reserves** are typically found in **inclined and folded sedimentary strata**, often in reservoirs at the crests of **anticlines.** Oil is frequently associated with water; being **lighter than water** (with a specific gravity of 0.8 to 0.98), oil accumulates in anticlines or fault traps above water. **Natural gas**, which is even lighter, typically occurs **above the oil layer.** When drilling an oil well, **gas is often encountered first, followed by oil.** However, the presence of gas does not always guarantee the presence of oil.

Petroleum Reserves in India:

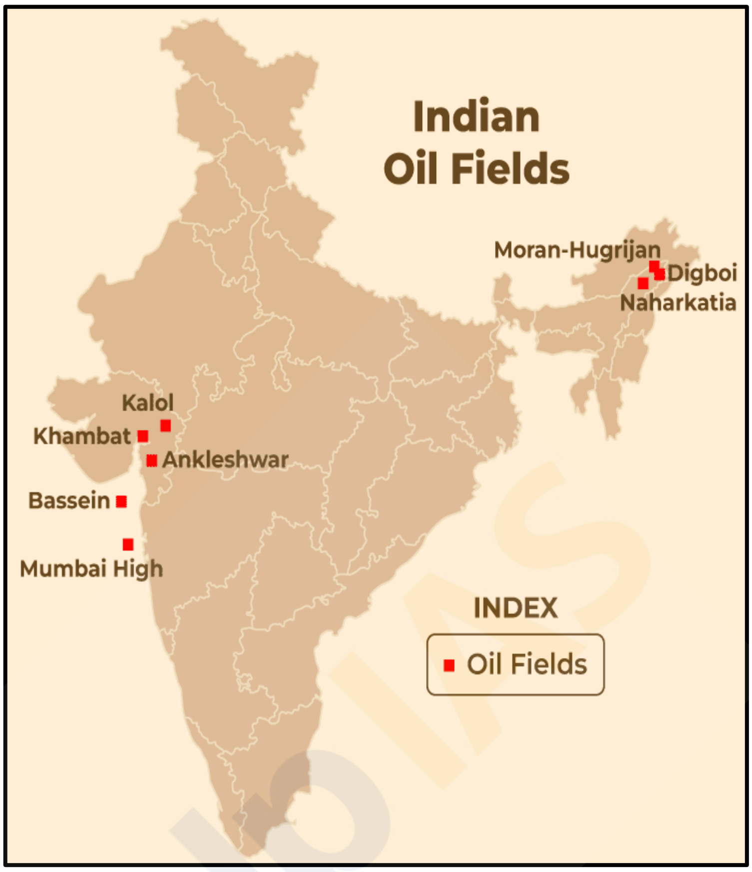


- In India, oil and natural gas are found in **sedimentary rocks**. About **14.1 lakh sq km**, or **42%** of the country's total area, is covered by sedimentary rocks. Of this, approximately **10 lakh sq km** consist of **marine basins** from the **Mesozoic and Tertiary periods**. Additionally, India's offshore areas, which contain **Mesozoic and Tertiary marine rocks**, cover about **2.5 lakh sq km** to a depth of 100 meters and another **0.7 lakh sq km** between 100 and 200 meters. Thus, the total **continental shelf** with probable oil-bearing rocks is estimated at **3.2 lakh sq km**.
- India's total sedimentary area, including both **onshore** and **offshore** regions, is divided into **27 basins**. **Geological** and **geophysical** studies have been conducted in **14 basins**, and **exploratory drilling** has been carried out in **9 basins**. The most productive oil regions in India are **Mumbai High**, the **Khambhat Gulf**, and **Assam**.

Oilfields in India:

- India is home to several oilfields spread across different regions, including the northeastern parts, western onshore fields, western offshore oilfields, and the eastern coast. These fields contribute to India's crude oil production and are vital to the nation's energy needs.

North-Eastern India:

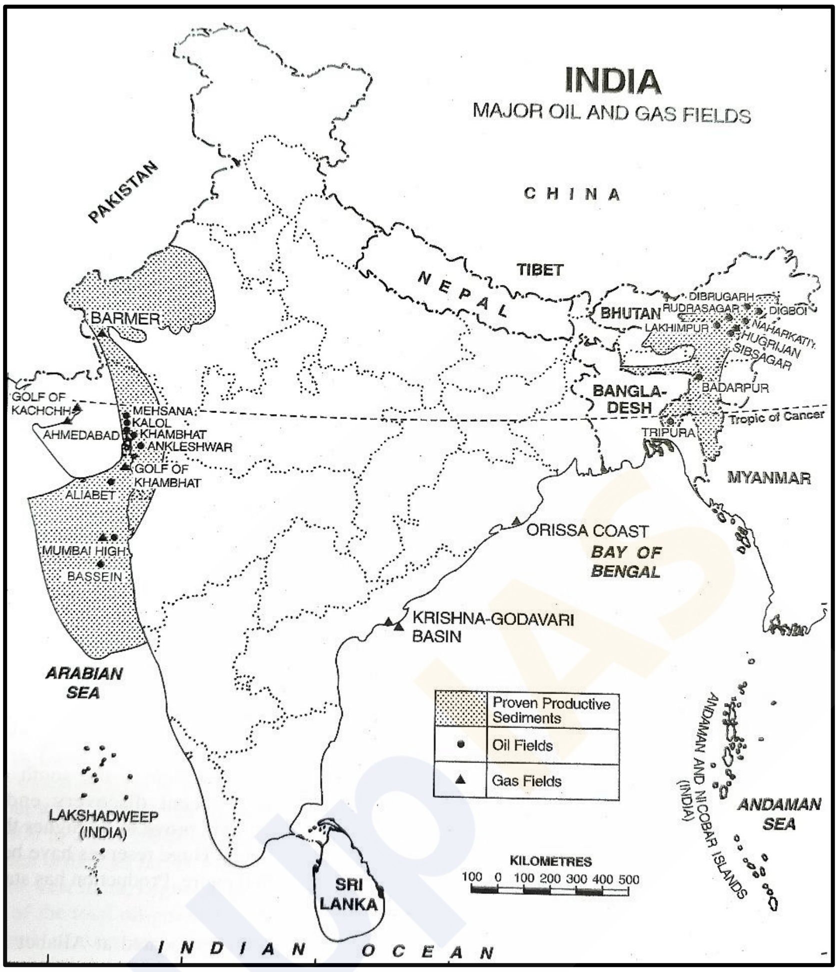


- The **northeastern region** of India is known for its rich oil reserves, primarily located in the **Brahmaputra valley** and its surrounding areas, including states like **Assam, Arunachal Pradesh, Nagaland, Tripura, Manipur, Mizoram,** and **Meghalaya**.
  - **Assam -** Assam is the oldest oil-producing state in India. The **Digboi oil field**, discovered in the late 19th century, is the oldest oilfield in the country. Most of the oil from this field is sent to the refinery at Digboi. Another notable field is **Naharkatiya**, discovered in 1953, which transports oil to refineries at Noonmati in Assam and Barauni in Bihar via pipelines.
  - **Arunachal Pradesh -** The state has oil reserves in areas like **Manabhaum, Kharsang,** and **Charai**, although production remains limited.
  - **Tripura -** In Tripura, oil reserves are found in **Manmumbhanga, Manu,** and **Ampi Bazar**.

Western India Onshore Oilfields:

- In **western India**, the state of **Gujarat** is particularly important due to its extensive oilfields, mainly concentrated around the **Gulf of Khambhat**. The primary oil belt stretches from **Surat** to **Amreli**, with several districts such as **Kachchh, Vadodara, Bharuch, Surat, Ahmedabad, Kheda, Mehsana** being major producers.
  - **Ankleshwar -** Discovered in **1958,** the **Ankleshwar oilfield** was the first major oil discovery in Gujarat, located about 80 km south of Vadodara and 160 km south of Khambhat. This field extends over 20 km and produces oil at depths between 1,000 to 1,200 meters. It has an annual production capacity of 2.8 million tonnes, with oil transported to refineries in Trombay and Koyali.
  - **Khambhat or Lunej field -** In 1958, test wells drilled at **Lunej**, near Ahmedabad, confirmed a commercially viable oil field. The production from this field is around 15 lakh tonnes annually, with gas production of 8-10 lakh cubic meters. The total reserves are estimated at **3 crore tonnes**.
  - **Ahmedabad and Kalol fields -** Located about 25 km northwest of Ahmedabad, this field, along with other nearby areas like **Nawgam, Kosamba, Mehsana, Sanand,** and **Kathana**, contains heavy crude oil trapped in coal seams.
  - **Rajasthan -** One of the largest onshore oil discoveries in India was made in **2004** in the **Barmer district** of Rajasthan. With the aid of innovative geological modeling, two important oilfields, **Sarswati** and **Rajeshwari**, were discovered, containing a total of 35 million tonnes of in-place oil reserves.

Western India Offshore Oilfields:



- **India's offshore oil exploration** has led to some significant discoveries, especially along the western coast.
  - **Mumbai High -** This is the most successful offshore oilfield discovered by **ONGC,** located about 176 km northwest of Mumbai on the continental shelf. The **Mumbai High** oilfield, discovered in 1974, covers an area of 2,500 sq km, with reserves estimated at **330 million tonnes** of oil and 37,000 million cubic meters of natural gas. Commercial production began in 1976, with oil extracted from a depth of over 1,400 meters using the **Sagar Samrat** platform.
  - **Bassein -** Located south of Mumbai High, the **Bassein oilfield** is a more recent discovery. It holds significant reserves found at a depth of 1,900 meters, with production steadily increasing.
  - **Aliabet -** Found on **Aliabet Island** in the **Gulf of Khambhat**, about 45 km off Bhavnagar, this field is another major offshore oil discovery with significant reserves.

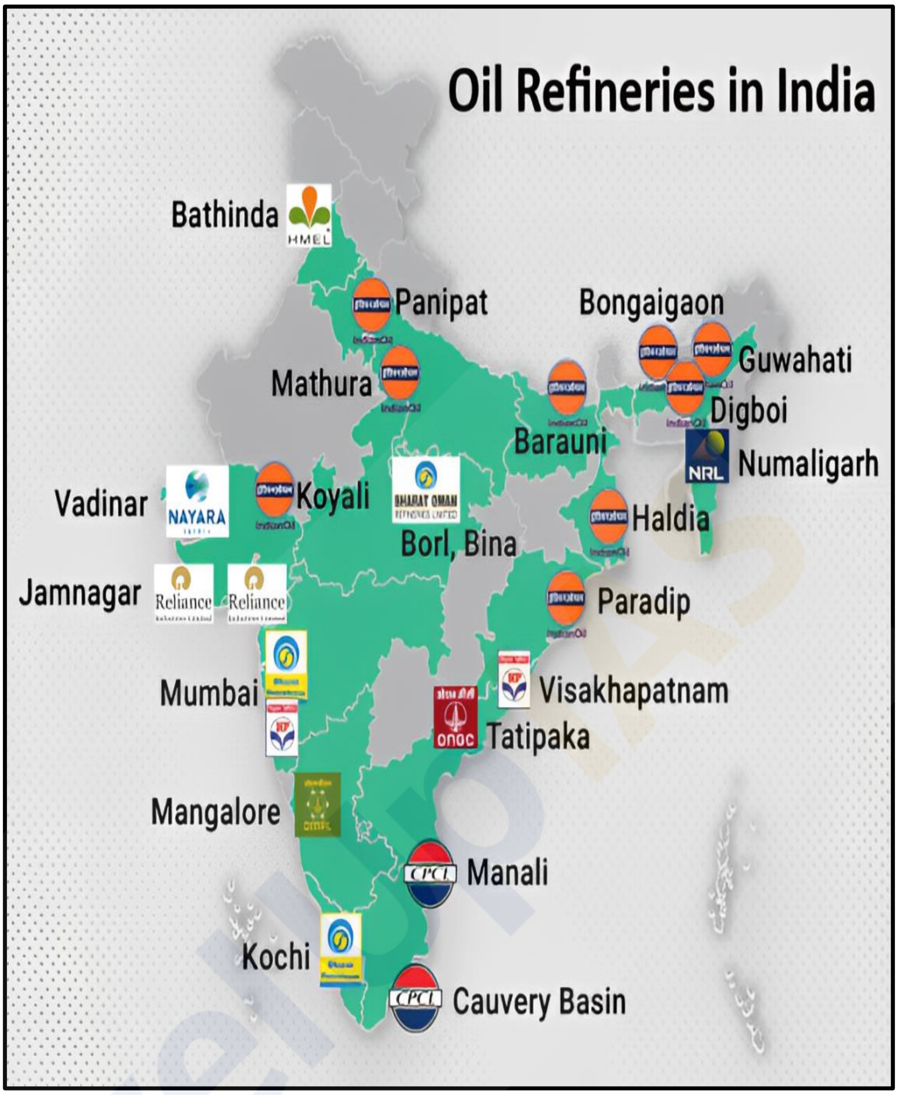
Eastern Coast:

- The **east coast** of India, particularly the basin and delta regions of the **Godavari, Krishna,** and **Cauvery rivers**, shows great potential for oil and gas production.
  - **Tamil Nadu -** The **Narimanam** and **Kovilappal** oilfields in the **Cauvery basin** have a production capacity of around 4 lakh tonnes of crude oil annually.
  - **Andhra Pradesh -** Although Andhra Pradesh produces less than **1%** of India's total crude oil, new oilfields have been discovered in the **Krishna-Godavari basin**.

Probable Areas for Future Exploration:

- India has several unexplored regions with promising potential for oil production, covering over one lakh sq km of sedimentary rock formations. These regions include:
  - **Himachal Pradesh -** Areas around **Jawalamukhi, Nurpur, Dharamsala,** and **Bilaspur**.
  - **Punjab -** Regions such as **Ludhiana, Hoshiarpur,** and **Dasua**.
  - **Tamil Nadu -** Offshore areas like the **Gulf of Mannar** off the **Tirunelveli coast**.
  - **Bay of Bengal -** The **offshore deepwater areas** and the marine delta regions of the **Mahanadi, Godavari, Krishna,** and **Cauvery rivers**.
  - **Andaman and Nicobar Islands -** Offshore areas around these islands, along with the sea stretch between **South Bengal** and the **Baleshwar coast**, are also considered promising.

Petroleum Refining:



- The Indian petroleum refining sector has evolved significantly since **crude oil was first discovered** and the **first refinery was set up at Digboi in 1901**. Until India's independence in 1947, Digboi was the sole refinery in the country, with a modest capacity of **0.50 million metric tonnes per annum (MMTPA)**. Post-independence, India saw the establishment of its first modern refinery at **Mumbai in 1954**, set up by **Esso**, followed by other major oil companies like **Burmah Shell** and **Caltex**, which built refineries at **Mumbai (BPCL)** and **Visakhapatnam (HPCL)**. The sector further expanded with the establishment of refineries by the **Government, Private Sector**, and **Joint Ventures**.
- India has experienced **remarkable growth** in the refining sector over the years. By **2001**, the country transitioned from a deficit in refining capacity to achieving **self-sufficiency** and emerging as a **major exporter** of high-quality petroleum products. Presently, India stands as a **global refining hub** with a capacity of **248.9 MMTPA**, making it the **fourth largest** refining country in the world, following the **United States, China,** and **Russia**. India's refining network comprises **23 refineries—18 in the Public Sector, 2 in Joint Ventures,** and **3 in the Private Sector**, all well distributed across the nation and connected via **cross-country pipelines**.

Evolution of Refinery Configurations:

- In the **1950s,** Indian refineries adopted **simple refinery configurations**, primarily focusing on **crude oil distillation** and basic processes like **naphtha, kerosene**, and **ATF treatment**, along with **catalytic reforming** to upgrade naphtha to petrol. These configurations lacked **secondary processing facilities**, resulting in **low energy recoveries** and reliance on **high sulfur fuel oil** as internal fuel.
- Refineries established in the **1960s** by Government undertakings primarily processed **low sulfur indigenous crude oils** from the **North East** and **Gujarat basin**. During this period, Indian **PSU refineries** began adopting **modern technologies** to align with global trends and meet product quality demands. **Secondary processing facilities**, such as **Fluid Catalytic Cracking (FCC) units**, were installed to convert **low-value streams** into **high-value middle distillates**.

- In the **late 1980s** and **early 1990s**, Indian refineries saw a major shift in configurations. There was an increased focus on the **maximisation of middle distillates** and the **stability of products**. Refinery configurations began to favor **Hydrocracking processes** alongside **FCC**, with many new refineries incorporating **hydrocrackers** as standard practice. The **first hydrocracker** in India was commissioned at the **Gujarat Refinery (IOCL) in 1993**. This phase also marked a significant rise in **investment requirements** due to the advanced nature of the processes involved.
- During the **late 1990s**, environmental considerations played a critical role in dictating refinery configurations. There was a need for **upgrading product quality**, leading to the development of **lead-free gasoline**, **low sulfur diesel**, and improved **fuel oil** for internal use. The demand for middle distillates continued to rise, prompting further enhancements in refinery setups.
- **Key technological additions during this period included:**
    - **Continuous Catalytic Reforming (CCR)**
    - **Hydrocracking units**
    - **Hydrotreating/Hydrodesulphurisation facilities** to produce **low sulfur fuels** for both external consumption and internal operations
    - **INDMAX Technology** for **LPG maximisation**
- These upgrades ensured that Indian refineries remained competitive globally while also meeting stricter **environmental regulations** and catering to the growing demand for **cleaner fuels**.

Significance of the Refinery Sector:

- **Energy Security -** Refineries are essential in ensuring **energy security** by processing crude oil into various petroleum products such as **petrol, diesel, kerosene, and LPG**, which meet the growing energy demands of the country. This helps reduce the dependence on external sources for refined products, making energy supply more secure and reliable.
- **Employment Generation -** The refinery sector contributes significantly to **employment generation**, both directly and indirectly. It creates job opportunities throughout the value chain, from **exploration and production** to refining, distribution, and retail operations. This enhances the livelihood of millions of individuals involved in the industry.
- **Infrastructure Development -** Establishing refineries leads to considerable **infrastructure development**, such as building pipelines, storage facilities, and transportation networks. These projects contribute to the **overall economic growth** of the regions where refineries are set up, fostering local development and improving logistical capacities.
- **Reducing Import Bill -** A strong domestic refinery sector helps **reduce India's import bill** by lowering the need for importing refined petroleum products. Processing crude oil domestically allows the country to meet a substantial portion of its demand for refined fuels, **saving foreign exchange** and reducing dependence on external markets.
- **Industrial Growth -** The refinery sector serves as a critical foundation for the growth of downstream industries. Industries such as **petrochemicals, fertilizers, and manufacturing** heavily rely on the products and by-products of refineries. This fosters industrial growth and strengthens the overall industrial ecosystem in the country.
- **Technological Advancements -** Refineries continuously invest in technological advancements, leading to the production of **cleaner fuels** and ensuring compliance with **environmental regulations**. For example, the production of **Bharat Stage VI (BS-VI) fuels**, which follow European emission norms, is crucial in reducing vehicular emissions and promoting sustainable practices in the sector.

- **Strategic Importance -** A **well-distributed refinery network** across the country mitigates risks related to **supply chain disruptions**. This ensures a stable and secure energy supply, safeguarding the country from any potential shortages due to geopolitical tensions or natural disasters.
- **Export Potential -** The refinery sector allows India to produce high-quality fuels that meet **international standards**. This positions the country as a player in the global energy market, contributing to **export revenues** by enabling the sale of refined products abroad.

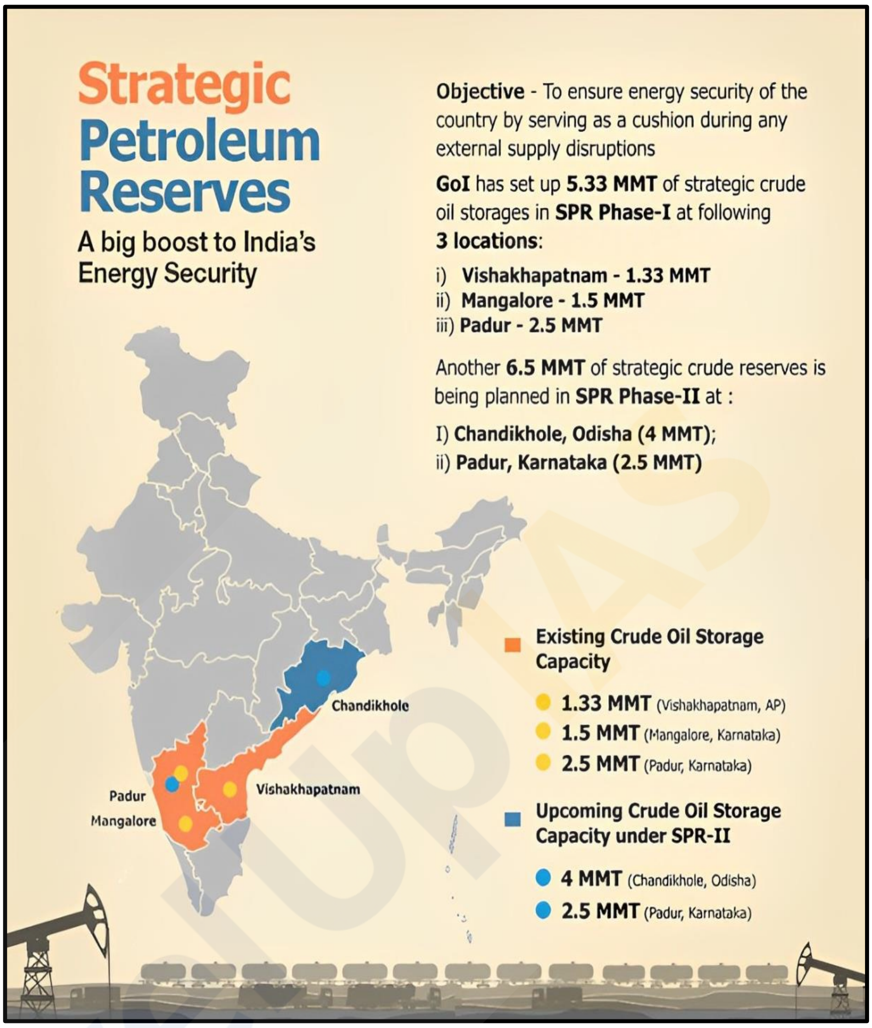
Challenges in the Refinery Sector:

- **Crude Oil Price Volatility -** The refinery sector is highly sensitive to **global crude oil price fluctuations**. Sudden and unpredictable changes in oil prices can affect the profitability of refineries and their ability to offer **competitive pricing** for refined products in both domestic and international markets.
- **Infrastructure Deficit -** An **inadequate infrastructure**, including insufficient pipelines, storage facilities, and transportation networks, creates logistical challenges. These deficits can disrupt the smooth operation of the supply chain, affecting the sector's overall efficiency.
- **Environmental Compliance -** Refineries must adhere to **stringent environmental regulations**, requiring significant investments in advanced technologies to reduce **greenhouse gas emissions** and produce cleaner fuels. Compliance with these regulations is essential but adds to operational costs and complexities.
- **Geopolitical Risks -** Global geopolitical issues, such as the **COVID-19 pandemic** and the **Russia-Ukraine conflict**, have significantly impacted the global oil market. These events have caused **supply shortages, price volatility,** and capacity reductions, disrupting the refining industry's operations.
- **Capital-Intensive Nature -** The refinery sector is **capital-intensive**, making it challenging to secure funding for new projects, expansions, and modernization efforts. Smaller players, in particular, struggle with accessing the necessary financial resources to stay competitive.
- **Policy Paralysis -** Frequent changes in government **policies and regulations** create uncertainty in the refinery sector. Stability in regulatory frameworks is crucial for **long-term planning and investments**, as uncertainty can delay decision-making and hinder project implementation.
- **Demand Fluctuations -** The refinery sector is subject to **demand fluctuations** due to economic uncertainties, global events, and shifts in consumer behavior. These fluctuations affect production levels and profitability, making it challenging for refineries to predict future demand trends.
- **Supply Chain Disruptions -** Private refineries in India have benefited from high global product prices, increasing their exports while curtailing supplies to the lower-priced domestic market. To prevent domestic refiners from prioritizing exports over meeting local demand, the **Indian government imposed a windfall tax on fuel exports**.

Way Forward:

- **Establishing a Refining-Petrochemical Hub -** India must continue on its path to becoming not just a refining hub but also a **refining-petrochemical hub**. This involves setting up **integrated projects** that can produce both fuels and petrochemicals, with the flexibility to adjust production based on market demand.
- **Large-Scale Projects -** The country should focus on building **large-scale refinery projects** that can compete in the global market and generate substantial **export revenues**. These projects will enhance India's position in the global energy market.
- **Strengthening Geopolitical Relationships -** India should leverage its **geopolitical relationships** with major oil-producing nations to secure **long-term crude oil supply**. This will ensure the stability of raw material supplies and enhance the security of energy resources needed for the country's refineries.

Strategic Petroleum Reserves:



- **Strategic Petroleum Reserves (SPR)** refer to **large stockpiles of crude oil maintained to address any crisis related to oil supply disruptions.** These disruptions can occur due to various reasons such as **natural disasters**, **wars**, or other unforeseen calamities. The purpose of these reserves is to ensure that countries have access to crude oil in case of emergency situations that could threaten **energy security**.
- Under the **International Energy Programme (I.E.P.) agreement**, member countries of the **International Energy Agency (IEA)** are required to maintain emergency oil reserves equivalent to **at least 90 days of their net oil imports**. This is a precautionary measure to ensure energy stability. In the event of a severe disruption in oil supply, the IEA can initiate a collective decision to release these reserves into the market to mitigate the impact.
- India became an **associate member of the IEA** in **2017** and has established its own strategic petroleum reserves to enhance energy security. Currently, India's strategic crude oil storage facilities are located at **Visakhapatnam (Andhra Pradesh)**, **Mangaluru (Karnataka)**, and **Padur (Karnataka)**. Additionally, the Indian government has approved the development of two more storage facilities at **Chandikhol (Odisha)** and **Padur (Karnataka)** to further expand the country's crude oil reserve capacity.
- The concept of **strategic petroleum reserves** first emerged in **1973 in the United States**, following the **OPEC oil crisis**. This crisis underscored the vulnerability of countries to external oil supply disruptions and led to the realization of the need for dedicated oil reserves. Today, SPR is considered essential for ensuring **energy security** in times of global or regional supply challenges.
- Among various methods of storing petroleum, **underground storage** is regarded as the most cost-effective and efficient. It reduces the need for large land areas, minimizes **evaporation losses**, and provides ease of discharging crude oil directly from ships due to the storage caverns being located **below sea level**. These factors make underground facilities a preferred option for storing petroleum products.
- The construction and management of India's strategic crude oil storage facilities are overseen by the **Indian Strategic Petroleum Reserves Limited (ISPRL)**, which is a wholly-owned subsidiary of the **Oil Industry Development Board (OIDB)** under the **Ministry of Petroleum & Natural Gas**. ISPRL is responsible for ensuring that the strategic reserves are maintained and can be effectively utilized during a crisis.

Pipelines:

- Pipelines are one of the most **convenient, efficient, and economical modes** of transporting liquids such as **petroleum, petroleum products, natural gas, water, and milk**. They are also capable of transporting solids, like **coal and iron ore**, after converting them into **slurry**. In recent years, the role of pipelines in India has expanded significantly, relieving the growing pressure on surface transport systems like **railways and roadways**.
- Pipelines have become essential in India for the **safe and affordable transportation of oil, gas, and petroleum products**. The **extensive network of pipelines** ensures a steady supply chain from production sites to refineries and industrial hubs across the nation. Once primarily used for **water transport**, pipelines now facilitate the movement of crucial resources, enabling **inland development** and reducing logistical challenges.
- The Indian government made a significant shift in **November 2002** with a policy that **deregulated the oil sector** and encouraged investment in **petroleum product pipelines**. This policy was based on the **common carrier principle**, which allowed multiple companies to use the same pipelines, promoting competition and efficiency. As a result, India's **underground pipeline network** has expanded rapidly, aiding in the seamless supply of oil, gas, and petroleum products.
- India is now among the **top five countries** in the world for **oil pipeline development**, with a considerable number of projects under construction or proposed. The country is currently constructing **1,630 km of oil transmission pipelines**, placing it **second globally** in ongoing pipeline projects, and it has **proposed an additional 1,194 km**, securing the **10th spot internationally**. Leading companies in oil pipeline development include **India's Numaligarh Refinery Limited**, alongside international players like **Iran's Ministry of Petroleum** and **China National Petroleum Corporation**.
- India is working on several large-scale projects, including the **Paradip-Numaligarh Crude Pipeline**, which is currently under construction, and the proposed **New Mundra-Panipat Oil Pipeline**, both of which will rank among the **longest pipelines globally**. These projects are set to play a crucial role in ensuring a stable supply of oil and gas across the country.

Advantages of Pipelines:

- They **consume less energy** than other forms of transportation, making them more efficient.
- **Transit losses** during pipeline transport are minimal compared to surface transportation.
- The **running cost** of materials through pipelines is significantly lower, up to **ten times less** than surface transportation.
- **Wastage of materials** during transport, such as slurry from coal and iron ore, is very low.
- Pipelines can be laid through **difficult terrains** and even **underwater**, which is not feasible for other transportation modes.
- They are not affected by **seasonal variations** or adverse weather conditions like **floods, snowfall, or rainfall**.
- Pipelines are **safe, accident-free**, and **environmentally friendly**, contributing to cleaner energy transport.
- They also spur **industrial development** in adjacent areas, as seen with the establishment of **fertilizer industries along gas pipelines** (e.g., **Jagdishpur, Shahjahanpur**).

Disadvantages of Pipelines:

- **Earthquakes** can cause distortions in pipelines, leading to **difficult-to-detect leakages** and cracks.
- The **initial cost** of laying pipelines is high, although the **maintenance cost** is relatively low.
- **Security threats** from terrorist organizations pose a risk to pipelines, particularly transboundary pipelines like the **TAPI pipeline**, which has faced threats from **Pakistan** and **Afghanistan**. Oil pipelines in the **northeast** have also been targeted by anti-nationalist elements.
- **Capacity limitations** are inherent once pipelines are laid; increasing capacity requires constructing new pipelines.
- Repairing pipelines, especially **detecting and fixing leakages**, can be extremely challenging.

Major Pipelines in India:

- The **pipeline network in India** forms a critical part of the country's infrastructure, supporting the transport of oil, gas, and other products. Some of the major pipelines include:
- **Naharkatia-Nunmati-Barauni Pipeline**: India's **first crude oil pipeline**, transporting oil from **Assam** to the **Barauni refinery**. It has several branch pipelines to increase its capacity.
- **Mumbai High-Mumbai Pipeline**: A **double pipeline** connecting offshore oil fields in the **Arabian Sea** to **Mumbai**, transporting crude oil and natural gas.
- **Ankleshwar-Koyali Pipeline**: Transports crude oil from Gujarat's **Ankleshwar oilfield** to the **Koyali refinery**.

- **Salaya-Koyali-Mathura Pipeline**: A **1,256 km pipeline** that supplies crude oil to refineries at **Koyali, Mathura, Panipat**, and **Jalandhar**, with an offshore terminal.
- **Hazira-Bijaypur-Jagdishpur Pipeline**: India's **longest gas pipeline**, transporting gas to power plants and fertilizer plants across several states.
- **Jamnagar-Loni LPG Pipeline**: The **world's longest LPG pipeline**, transporting LPG from **Jamnagar** to **Loni** near Delhi, meeting the demands of several states.

Natural Gas:

- **Natural Gas**, also referred to as **fossil gas** or simply **gas**, is a naturally occurring **hydrocarbon gas mixture**. Its primary component is **methane**, but it commonly contains other higher alkanes and small amounts of gases such as **carbon dioxide, nitrogen, hydrogen sulfide**, and **helium**. It is also classified as a **fossil fuel**, often found in the same geological formations as **petroleum**.
- In some cases, the pressure of **natural gas** forces oil to the surface, a form known as **associated gas** or **wet gas**. When reservoirs contain only gas and no oil, this is referred to as **non-associated gas** or **dry gas**. Gases containing substantial quantities of **hydrogen sulfide** or other sulfur compounds are classified as **sour gas**. Conversely, **coalbed methane**, which lacks hydrogen sulfide, is termed **sweet gas**.
- On the market, **natural gas** is usually bought and sold based on its **calorific value** rather than by volume. In practice, transactions are often measured in **MMBTUs (millions of British thermal units)**, equivalent to roughly **1,000 cubic feet of natural gas**. Despite being **odorless** and **colorless** in its natural state, gas for domestic use is infused with **mercaptan compounds** for safety and leak detection, giving it the familiar sour smell.
- **History of Natural Gas in India**
- India's commercial production of **natural gas** began relatively late, starting in **1961**. The natural gas industry in India took off in the **1960s** with the discovery of gas fields in **Assam** and **Gujarat**. The establishment of **GAIL** in **1984** marked a turning point, leading to increased focus on natural gas production.

Formation of Natural Gas

- **Natural gas** forms over millions of years through the decomposition of **plant** and **animal matter** under intense **heat** and **pressure** deep beneath the Earth's surface. The energy that these plants originally captured from the sun is stored in **chemical bonds** within the gas.

Uses of Natural Gas:

- In the **19th** and **early 20th** centuries, **natural gas** was primarily used for **street** and **household lighting**. Today, its applications have expanded significantly:
    - It powers **turbines** for wind and solar energy generation.
    - It is widely used as a **domestic fuel** for **stoves, heaters, ovens,** and **boilers**.
    - **Compressed Natural Gas (CNG)**, stored under high pressure, is a cheap, **environment-friendly** alternative for transportation, especially in **low-load vehicles** requiring high fuel efficiency.
    - **Liquefied Natural Gas (LNG)** is used to fuel **off-road trucks** and **trains**.
    - **CNG** also powers **buses** and **commercial automotive fleets**.
    - It serves as a critical ingredient in **dyes**, **inks**, and **rubber compounding operations**.
    - **Ammonia**, used in producing chemicals like **hydrogen cyanide, nitric acid**, and **fertilizers**, is manufactured using hydrogen derived from methane.

Importance of Natural Gas:

- **Natural gas** plays a crucial role in India's **fertilizer production**, with about **40%** of the total gas consumed for this purpose. Another **30%** is used in **power generation**, and **10%** goes towards **LPG** production. The growth of **natural gas** has complemented the development of these sectors, particularly after **1971**, when there was a significant increase in gas production.

- Although **gas power stations** contribute about **10%** of India's electricity generation, many are currently idle due to a **lack of feedstock**. Despite the country's power crisis, existing plants operate below capacity, relying on **expensive imported LNG**.
- India's **oil reserves** are insufficient to meet its growing energy needs, a situation exacerbated by **policy delays** that prolong project timelines. To avoid external shocks, India needs to diversify its **energy basket** by incorporating **alternative fuels**. Globally, **natural gas** accounts for about **25%** of energy consumption, but in India, it constitutes only **6%**, with **crude oil** and **coal** dominating. The government aims to increase natural gas's share to **15% by 2030**.

Distribution of Natural Gas in India

- **India's economically viable natural gas reserves are estimated at:**
  - **541 BCM** onshore, mainly in **Assam** and **Gujarat**.
  - **190 BCM** offshore in the **Bay of Cambay**.
  - **190 BCM** in **Bombay High**.
- Recently, a massive reserve of **400 BCM** was reported in the **Tripura Basin**. In addition, **72 BCM** is located in the **Rava structure**, and a substantial reserve has been identified around the **Andaman and Nicobar Islands**. Remote sensing estimates place this reserve at **1,700 BCM**, which could supply India's needs for over **100 years**, though its **economic viability** has yet to be established.

Major Discoveries of Natural Gas:

- **1988-89 -** Significant oil discoveries at **Cauvery offshore** and **Nanda** in the **Khambhat basin**, with gas found in **Tanot**, **Jaiselmer basin**, Rajasthan.
- **1988 -** Production from the **South Bassein Gas Field** commenced in September.
- **1989-90 -** Gas structures were found in **Adiyakkamanglam** (Tamil Nadu), **Andada** (Gujarat), **Khovaghat** (Assam), **Lingla** (Andhra Pradesh), **Mumbai Offshore**, and **Kachchh offshore**.
- **2002 - Reliance Industries** made a landmark discovery in the **Krishna-Godavari basin**, considered the world's largest gas find in that year. The field holds reserves of **14 trillion cubic feet** and yields **60-80 million cubic meters** of gas daily.
- **2003 -** Gas was discovered alongside crude oil in **Barmer district**, Rajasthan.
- **2004 -** Another gas discovery off the **Orissa coast** in the **Bay of Bengal**.
- **2005 - ONGC** found hydrocarbons in the shallow waters of the **Krishna-Godavari basin**, southwest of the **Rava field** discovered in **1987**. This new find is located **12 km** from the **Amalpuram coast**.

Nuclear Energy:

- **Nuclear energy** is derived primarily from **uranium** and **thorium**. India possesses **vast untapped uranium resources**, making it crucial to utilize these resources to address the current **power shortages** and **energy crisis**. At present, nuclear power contributes a little over **3%** to India's total power generation, but it holds significant potential for future expansion. India has successfully developed the **technology** required for nuclear power generation, enabling it to design, construct, commission, and operate nuclear power stations independently.

Major Atomic Minerals:

Thorium:

- **Thorium** is primarily obtained from **monazite**, a mineral that contains about **10% Thoria** (thorium oxide) and **0.3% Urania** (uranium oxide). Another mineral that contains thorium is **thorianite**. India boasts the largest deposits of thorium globally and is focused on progressing to the **third stage** of nuclear fuel consumption, which relies heavily on thorium, to achieve **self-reliance** in nuclear fuel supply.

Uranium:

- **Uranium** is a **silvery-gray metallic** element, denoted by the chemical symbol **U** and atomic number **92**. It is radioactive and naturally formed only during **supernova explosions**. Among its isotopes, **Uranium-238 (99.27%)** and **Uranium-235 (0.72%)** are notable, with only Uranium-235 being **fissile**, meaning it can sustain a nuclear chain reaction.
- India's total **uranium reserves** are estimated at **30,480 tonnes**, and it currently produces about **2 percent** of the world's uranium supply. Most uranium deposits are found in **crystalline rocks**, with approximately **70 percent** of the deposits located in **Jharkhand**. Other notable locations include the **Singhbhum** and **Hazaribagh districts of Jharkhand**, **Gaya district of Bihar**, and sedimentary rocks in **Saharanpur district of Uttar Pradesh**.

Other Economically Viable Atomic Minerals:

Monazite:

- Monazite sands, which occur along the **east and west coasts of India** and in parts of **Bihar**, have the largest concentration on the **Kerala coast**. It is estimated that monazite contains over **15,200 tonnes** of uranium.

Ilmenite:

- **Ilmenite** deposits are found primarily in **Jharkhand**, contributing to the country's atomic mineral resources.

Beryllium:

- **Beryllium oxide** is utilized as a **moderator** in nuclear reactors. India possesses adequate reserves of beryllium, which is primarily found in **Rajasthan, Jharkhand, Andhra Pradesh,** and **Tamil Nadu**.

Zirconium:

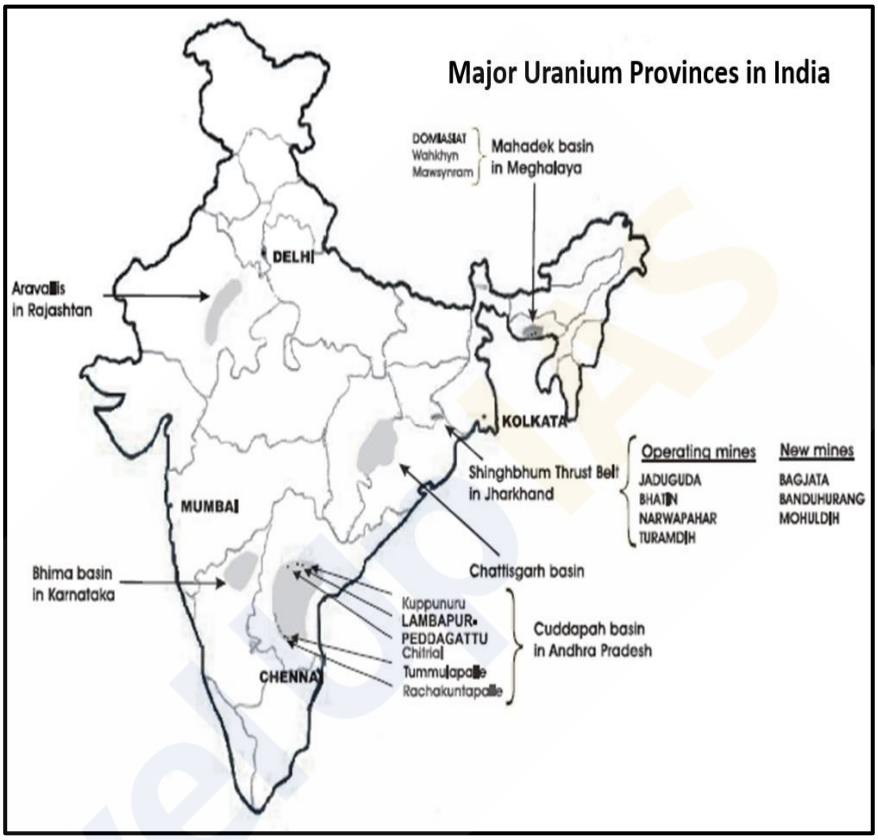
- **Zirconium** occurs along the **Kerala coast** and in the **alluvial rocks** of **Ranchi** and **Hazaribagh districts** of Jharkhand.

Lithium:

- **Lithium** is a lightweight metal found in **lepidolite** and **spodumene**. Lepidolite is distributed throughout the **mica belts** of **Jharkhand, Madhya Pradesh, Rajasthan**, and the **Bastar region** of **Chhattisgarh**.

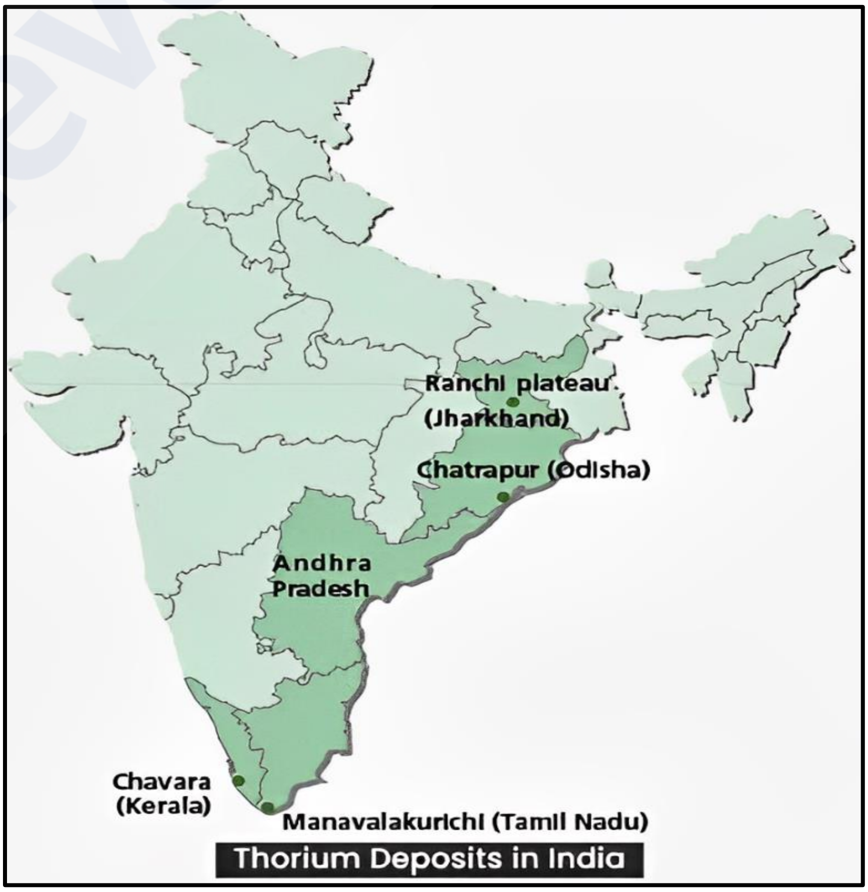
Uranium Mining in India:

- India currently lacks significant uranium reserves and meets its needs primarily through imports from countries such as **Russia, Kazakhstan,** and **France**. Efforts are underway to import uranium from **Australia** and **Canada**, despite challenges related to **nuclear proliferation**. Recent discoveries in **Andhra Pradesh** and **Telangana** between the **Seshachalam forest** and **Srisailam** are promising.
- **Notable Uranium Mines:**
    - **Jharkhand -** Notable mines include **Jaduguda**, **Bhatin**, **Narwapahar**, **Bagjata**, **Turamdih**, **Banduhurang**, and **Mohuldih**.
    - **Meghalaya -** Includes mines at **Kylleng**, **Pyndeng-Shahiong (Domiasiat)**, **Mawthabah**, and **Wakhym**.
    - **Andhra Pradesh -** The **Tummalapalle** mine has substantial reserves, potentially increasing to become the world's largest.



Thorium Reserves in India:

- India possesses **significant thorium reserves**, primarily concentrated in the **southern coastal regions**, particularly in **Kerala** and **Tamil Nadu**. These reserves are estimated to be among the **largest in the world**, offering a potential alternative fuel source for **nuclear energy**.
- **Thorium** is a naturally occurring radioactive element that can be utilized as a nuclear fuel. Notably, it has the potential to **produce more energy than uranium** while generating **less radioactive waste**. Recognizing this potential, India has been actively researching and developing **thorium-based nuclear reactors** as part of its **long-term energy strategy**. The goal is to harness the vast thorium reserves to ensure energy security.



- However, it is important to note that the development of thorium-based nuclear technology is still in its **early stages**. There are significant **technical** and **economic challenges** that must be addressed before this technology can be **widely implemented**.

Historical Development:

- The **nuclear power program** in India was initiated in the **1940s**, with the incorporation of the **Tata Atomic Research Commission** in August **1948**. However, substantial progress began with the establishment of the **Atomic Energy Institution** at **Trombay** in **1954**, later renamed the **Bhabha Atomic Research Centre (BARC)** in **1967**. The first nuclear power station, with a capacity of **320 MW**, was established at **Tarapur** near **Mumbai** in **1969**. Additional reactors were subsequently installed in various locations, including **Rawatbhata** (300 MW) in **Rajasthan**, **Kalpakkam** (440 MW) in **Tamil Nadu**, and **Narora** in **Uttar Pradesh**. Other significant sites include **Kaiga** in **Karnataka** and **Kakarapara** in **Gujarat**. Future projects are proposed at locations such as **Kumharia** (Haryana), **Bargi** (Madhya Pradesh), **Haripur** (West Bengal), **Jailapur** (Maharashtra), **Mithi Viridi** (Gujarat), and **Kovvada** (Andhra Pradesh).

Importance of Nuclear Energy:

- **Availability of Thorium -** India is a global leader in the availability of **thorium**, a potential nuclear fuel for the future. With abundant thorium reserves, India has the opportunity to become the first nation to achieve a **fossil fuel-free** status.
- **Economic Benefits -** Utilizing nuclear energy could save India approximately **$100 billion annually** currently spent on importing petroleum and coal. Furthermore, **nuclear power** offers a **stable and reliable** energy source, unlike solar and wind energy, which are weather-dependent.
- **Cost-Effectiveness -** Nuclear power plants are generally **cheaper to operate** than coal or gas facilities. Estimates suggest that, even when including the costs associated with managing radioactive fuel and waste disposal, nuclear plants operate at **33-50%** of the cost of coal plants and **20-25%** of the cost of gas combined-cycle plants.

Three-Stage Nuclear Power Program:

- To address the limited availability of fossil fuels in India, a strategic **three-stage nuclear power program** was formulated by **Dr. Homi Bhabha**. This program aims to link the fuel cycles of **Pressurized Heavy Water Reactors (PHWR)** and **Fast Breeder Reactors (FBR)** to utilize India's limited **uranium** and abundant **thorium** resources effectively.
- **Stage 1: Pressurized Heavy Water Reactor**
  - This stage involves the construction of **Natural Uranium Heavy Water moderated** and cooled **Pressurized Heavy Water Reactors (PHWRs)**. The **spent fuel** from these reactors is reprocessed to obtain **plutonium**.
- **Stage 2: Fast Breeder Reactor**
  - In the second stage, **Fast Breeder Reactors (FBRs)** are built, fueled by the **plutonium** produced in stage one. These reactors also breed **U-233** from thorium.
- **Stage 3: Breeder Reactor**
  - The final stage consists of **power reactors** that utilize **U-233/Thorium** as fuel, produced during stage two.

Benefits of Nuclear Energy

- **National Security -** Ensures safety and non-proliferation standards globally while supporting a resilient electrical grid.
- **Combatting Climate Change -** Provides a significant amount of **carbon-free** electricity, crucial for environmental protection.
- **Technological Leadership -** The U.S. pioneered nuclear energy, and continued leadership can meet growing global clean energy demands.
- **Reliability -** Nuclear power delivers around-the-clock electricity, essential for national prosperity.
- **Job Creation -** Provides over **100,000 well-paid jobs** and contributes significantly to local economies through tax revenues.
- **Air Quality Protection -** Operates without emitting nitrogen oxides, sulfur dioxide, and other harmful pollutants.
- **International Development -** Assists developing nations in meeting their **sustainable development goals**.

Challenges to Nuclear Energy Adoption:

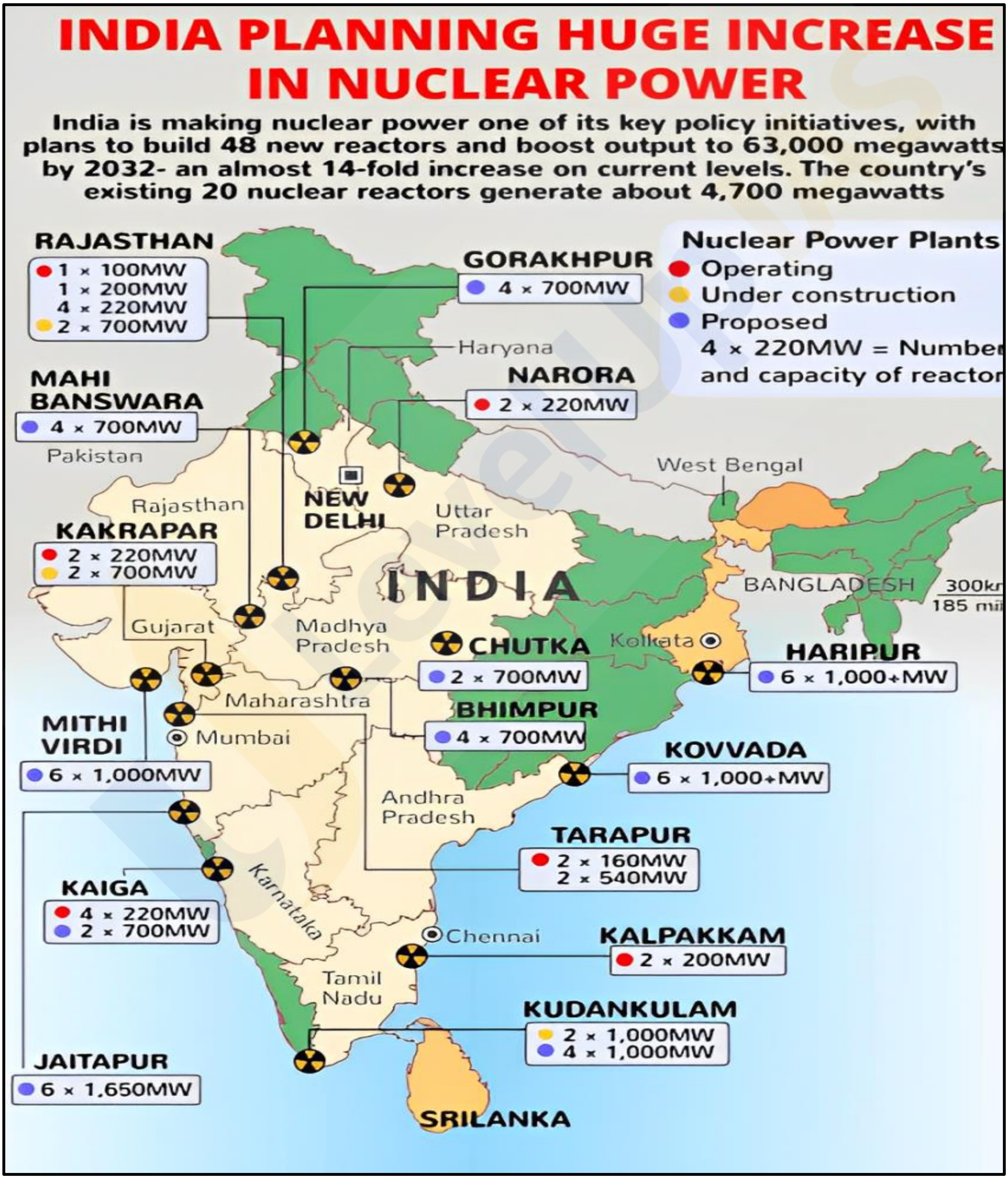
- **Capital Intensity -** Nuclear power plants require substantial investment, and recent projects have encountered significant **cost overruns**. A notable example is the **V.C. Summer project in South Carolina,** where costs escalated to over **$9 billion**, leading to project abandonment.
- **Insufficient Installed Capacity -** While the Atomic Energy Commission projected **650 GW** of capacity by **2050**, the current installed capacity stands at only **6.78 GW**.
- **Lack of Public Funding -** Nuclear energy has not received the same level of subsidies as fossil fuels and renewables, limiting its competitive edge.
- **Land Acquisition Issues -** The selection and acquisition of land for nuclear power plants pose significant challenges, as seen with projects like **Kudankulam** in **Tamil Nadu** and **Kovvada** in **Andhra Pradesh**, which faced delays.
- **Impact of Climate Change -** Increasing temperatures and changing weather patterns raise the risk of **nuclear reactor accidents**. Many plants rely on nearby water sources for cooling, which may become scarce.
- **Deployment Scale -** Nuclear power may not be scalable enough to mitigate India's **carbon emissions** effectively.
- **Nuclear Waste -** The production of nuclear waste, which poses health risks and environmental concerns, is a significant drawback, particularly in densely populated areas.

Operational Nuclear Power Plants in India:

Name of Nuclear Power Station	Location	Operator	Capacity (MW)
Kakrapar Atomic Power Station - 1993	Gujarat	NPCIL	440
(Kalpakkam) Madras Atomic Power Station - 1984	Tamil Nadu	NPCIL	440
Narora Atomic Power Station - 1991	Uttar Pradesh	NPCIL	440
Kaiga Nuclear Power Plant - 2000	Karnataka	NPCIL	880
Rajasthan Atomic Power Station - 1973	Rajasthan	NPCIL	1,180
Tarapur Atomic Power Station - 1969	Maharashtra	NPCIL	1,400
Kudankulam Nuclear Power Plant - 2013	Tamil Nadu	NPCIL	2,000

Under Construction Nuclear Power Plants in India:

Name of Nuclear Power Station	Location	Operator	Capacity (MW)
Madras (Kalpakkam)	Tamil Nadu	BHAVINI	500
Rajasthan Unit 7 and 8	Rajasthan	NPCIL	1,400
Kakrapar Unit 3 and 4	Gujarat	NPCIL	1,400
Kudankulam Unit 3 and 4	Tamil Nadu	NPCIL	2,000



Non-Conventional Sources of Energy:

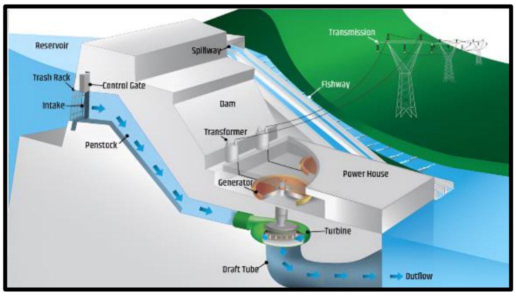
Non-conventional sources of energy are essential alternatives to traditional fossil fuels, offering **sustainable** and **environmentally friendly** options for meeting the world's energy needs. These sources include **solar energy**, harnessed through **photovoltaic cells** or **solar thermal systems**, which convert sunlight into electricity or heat. **Wind energy** is captured using **turbines** that convert the **kinetic energy** of wind into electrical power. **Hydropower** utilizes the energy of **flowing water** to generate electricity, often through **dams** or **run-of-river systems**. **Geothermal energy** exploits heat from the Earth's interior to produce electricity or provide direct heating. **Biomass** involves converting **organic materials**, such as agricultural residues or wood, into energy through combustion or biochemical processes. Additionally, **ocean energy**, including **tidal** and **wave power**, taps into the energy of ocean movements. These non-conventional sources are crucial for reducing **greenhouse gas emissions**, mitigating **climate change**, and decreasing dependency on finite fossil fuels, thus contributing to a more **sustainable** and **resilient energy future**.

Hydroelectricity:

- **Hydroelectricity**, also known as **hydroelectric power** or **hydroelectric energy,** is a **renewable energy source** that generates electricity by harnessing the **power of moving water.** This is typically achieved by directing water from higher to lower elevations, where the **flow's kinetic energy turns a turbine.** This **mechanical energy** is then **converted into electrical energy** via a generator. The electricity generated is subsequently transmitted to a **substation**, where transformers increase the voltage before it is distributed through the **electricity grid**.
- Hydropower is regarded as one of the **cheapest and cleanest** forms of energy. However, large-scale projects like the **Tehri** and **Narmada** dams have raised **environmental and social concerns** due to their impact on ecosystems and displaced communities. In contrast, **small hydropower** projects tend to avoid these issues, making them a more sustainable option.
- India, which currently has **197 hydroelectric power plants**, has a long history with hydropower. The first instance of electricity generation in India occurred in **1897 in Darjeeling**, followed by the commissioning of a **Hydro Power station at Sivasamudram in Karnataka** in 1902.

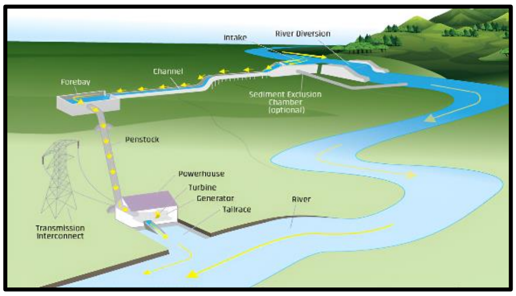
Types of Hydropower Facilities:

Hydropower plants are generally classified into three main types: **impoundment**, **diversion**, and **pumped storage**. Some of these facilities utilize dams, while others do not.



Impoundment Facility:

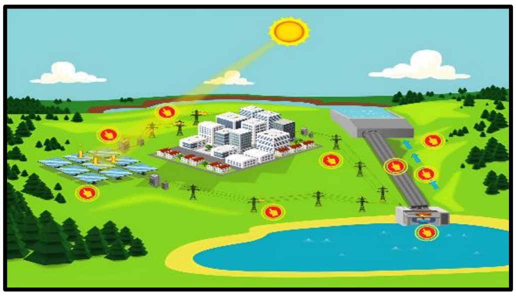
- This is the **most common type** of hydropower plant. An impoundment facility uses a **dam** to store river water in a reservoir. When **water is released from the reservoir, it flows through a turbine, spinning it, which activates a generator to produce electricity.**



Diversion Facility:

- Also known as a **run-of-river facility**, this type channels a portion of a **river's flow through a canal** or **penstock to drive a turbine.** It may not require a dam, making it less intrusive to the natural flow of the river.

Pumped Storage Facility:



- This facility acts like a **battery**, storing the electricity generated from other sources, such as **solar**, **wind**, or **nuclear energy**, for later use. When electricity demand is low, water is pumped from a **lower reservoir** to an **upper reservoir**. During peak demand, the stored water is released back into the lower reservoir, spinning a turbine to generate electricity.

Classification and Management of Hydro Power Projects in India:

- Hydropower projects in India are categorized into different segments based on their **installed capacity**. They are generally divided into **small** and **large** hydropower projects:
    - **Micro Hydropower -** Upto **100 KW** capacity.
    - **Mini Hydropower -** Ranging from **101 KW to 2 MW**.
    - **Small Hydropower -** Between **2 MW and 25 MW**.
    - **Mega Hydropower -** Projects with an installed capacity of **500 MW or more**.
- Initially, the management of hydro power, including small hydro projects, was under the purview of the **Ministry of Power** until 1989. In that year, the responsibility for plants with a capacity of **3 MW and below** was transferred to the **Ministry of New and Renewable Energy (MNRE)**. Subsequently, in **November 1999**, the MNRE was entrusted with the responsibility of managing hydro plants with a capacity of **up to 25 MW**.

Advantages of Hydropower:

- **Renewable and Non-Consumptive Energy Source -** Hydropower is a **renewable source of energy** because it uses the energy of flowing water without depleting the resource. The water remains available for other purposes such as irrigation, drinking, and industrial use after passing through the turbines.
- **Minimal Recurring Costs -** Unlike fossil fuel-based energy sources, **hydropower does not rely on consumables like coal, oil, or gas**. This leads to minimal operational costs, as the primary resource—water—is naturally replenished. This lack of consumable requirements ensures that hydropower is **economically viable in the long term**, with **no significant recurring expenses**.
- **Cost-Effective Energy Generation -** Hydropower is **cheaper** compared to electricity generated from **coal and gas-fired plants**. Moreover, it **reduces financial losses caused by frequency fluctuations** in electrical grids, providing more stable and reliable electricity. As hydropower does not depend on fossil fuels, it is **inflation-free** and less vulnerable to market fluctuations in fuel prices.
- **Efficient for Peak Load Demand -** One of the unique features of hydropower stations is their ability to **start and stop operations quickly**, making them ideal for meeting **peak electricity demands**. This **flexibility in operation** is crucial for managing grid stability, especially during sudden increases in electricity consumption.
- **Complementary to Thermal Power -** Hydropower stations complement thermal power plants in mixed energy systems. The **balanced combination of hydro and thermal energy** allows for **optimal utilization** of the available capacity, improving the overall efficiency of the energy grid.

- **Seasonal Load Adaptability -** Hydropower generation patterns often align well with the seasonal load demands of regional grids. During the **summer and monsoon seasons**, when agricultural activities are high and water availability is abundant, hydropower plants can generate more electricity to meet the **increased demand**. In contrast, during **winter**, thermal power plants operate at base load while hydropower stations manage the **peak load requirements**, ensuring a steady and reliable supply.

Disadvantages of Hydropower:

- **High Capital Investment -** Hydropower generation is **capital-intensive**, requiring a large upfront investment for the construction of dams, powerhouses, and other infrastructure. The initial costs can be prohibitive, although the long-term operational costs are low.
- **Environmental Impact -** Hydropower projects, especially large dams, are typically located in **hilly areas**, where forest cover is more extensive. As a result, the **diversion of forest land** for dam construction can have a detrimental impact on local ecosystems. Furthermore, large reservoirs created by dams lead to the **submergence of land**, causing **loss of flora and fauna**.
- **Displacement of Communities -** One of the most challenging aspects of hydropower projects is the **displacement of local populations**. The creation of reservoirs often leads to the **submergence of agricultural land**, forcing farmers and residents to relocate. This displacement can lead to social, economic, and cultural disruptions in affected communities.
- **Geographical Limitations -** Hydropower dams can only be constructed in **specific locations** where suitable river flows and geographic features exist. This limits the widespread application of hydropower, especially in regions without adequate water resources or favourable topography.
- **Submergence of Agricultural Land -** Hydropower projects often result in the **submergence of large tracts of agricultural land**, which can have a significant impact on local food production and livelihood. The loss of fertile land for farming is a critical concern, particularly in densely populated regions dependent on agriculture.

Hydropower Scenario in India:

- India ranks **5th globally in terms of installed hydroelectric power capacity**. As of **31 March 2020**, India's **installed utility-scale hydroelectric capacity** was **46,000 MW**, accounting for **12.3%** of the country's total utility power generation capacity. Additionally, smaller hydroelectric power units with a total capacity of **4,683 MW** (1.3% of total utility power generation capacity) have been installed.
- India's total hydroelectric power potential is estimated at **148,700 MW** at a **60% load factor**. This includes an additional potential of **6,780 MW** from small hydro schemes (projects with capacities less than 25 MW). Small Hydropower (SHP) potential is further assessed to be **21,135 MW** from **7,135 sites**, as evaluated by the **Alternate Hydro Energy Centre (AHEC) of IIT Roorkee** in its 2016 Small Hydro Database.
- The **northern and northeastern regions** hold the **majority of India's hydropower potential. Arunachal Pradesh** leads with **47 GW** of unexploited potential, followed by **Uttarakhand** with **12 GW**. The potential remains largely untapped along the **Indus, Ganges, and Brahmaputra River systems**. India also has over **90 GW of pumped storage potential**, with **63 sites** recognized in national energy policies for their grid services.
- In the **fiscal year 2019–20**, India generated **156 TWh** from hydroelectric power (excluding small hydro) with an **average capacity factor of 38.71%**.

- The **public sector** dominates hydropower production, contributing **92.5%** of total generation. However, the **private sector** is expected to play a growing role, especially with the development of projects in the **Himalayan** and **northeastern** regions.
- India also imports surplus hydroelectric power from **Bhutan**.

Challenges faced by India in Hydropower Generation:

- **Underdeveloped Hydropower in Central India -** Major hydropower potential from river basins like the **Godavari**, **Mahanadi**, **Nagavali**, **Vamsadhara**, and **Narmada** remains underdeveloped due to opposition from **tribal populations**.
- **Declining Share in Electricity Mix -** Hydropower's share in India's electricity mix has decreased over time, contributing around **10% of total generation**, while **thermal generation** accounts for **80%**.
- **Project Delays -** Several hydropower projects face delays due to complex planning, prolonged **land acquisition**, lack of **infrastructure**, and insufficient **financing**. Other factors include **contractual conflicts**, **environmental litigations**, and **local disturbances**.
- **Conflict Between States -** As water is a **state subject**, conflicts between **riparian states** further delay projects. A notable example is the **Subansiri Hydroelectric Project**.
- **Clearance and Environmental Issues -** Hydropower projects require various **environmental clearances**, which often cause delays. Issues like **environmental flow (e-flow)**, **eco-sensitive zones**, and impacts on **flora and fauna** are now more clearly defined.
- **Techno-Economic Clearance (TEC) -** Unlike **thermal projects**, hydroelectric projects require **TEC** from the **Central Electricity Authority (CEA)** if the capital expenditure exceeds ₹1,000 crore, adding to delays. Site-specific changes during construction also need **CWC approval**, further extending timelines.
- **Resettlement and Rehabilitation (R&R) Issues -** Resettlement and rehabilitation for displaced populations is a sensitive issue and adds substantial costs to projects. Delays in this process can lead to cost and time overruns.
- **Difficult Accessibility -** Hydropower projects are often located in **remote and inaccessible** areas, requiring the development of **roads and bridges**, which also boost regional development. The Government of India provides **budgetary support** for such infrastructure development.
- **Debt and Tariff Structure -** Hydropower projects have a **70:30 debt-equity ratio**, with the tariff designed to recover debt within the first **12 years**, which can make hydroelectric energy **unviable** due to the **front-loaded tariff** structure. To address this, the government has extended the debt repayment period to **18 years** and the project life to **40 years**, along with an annual **2% tariff escalation**.
- **Cost Overruns and Conflict Resolution -** Unforeseen factors like **geological surprises**, **R&R issues**, and **environmental factors** often result in **cost and time overruns**. Projects face **litigation** and **contractual conflicts**, which further delay implementation. A robust mechanism is required for the **quick resolution** of contractual disputes.

Way Forward:

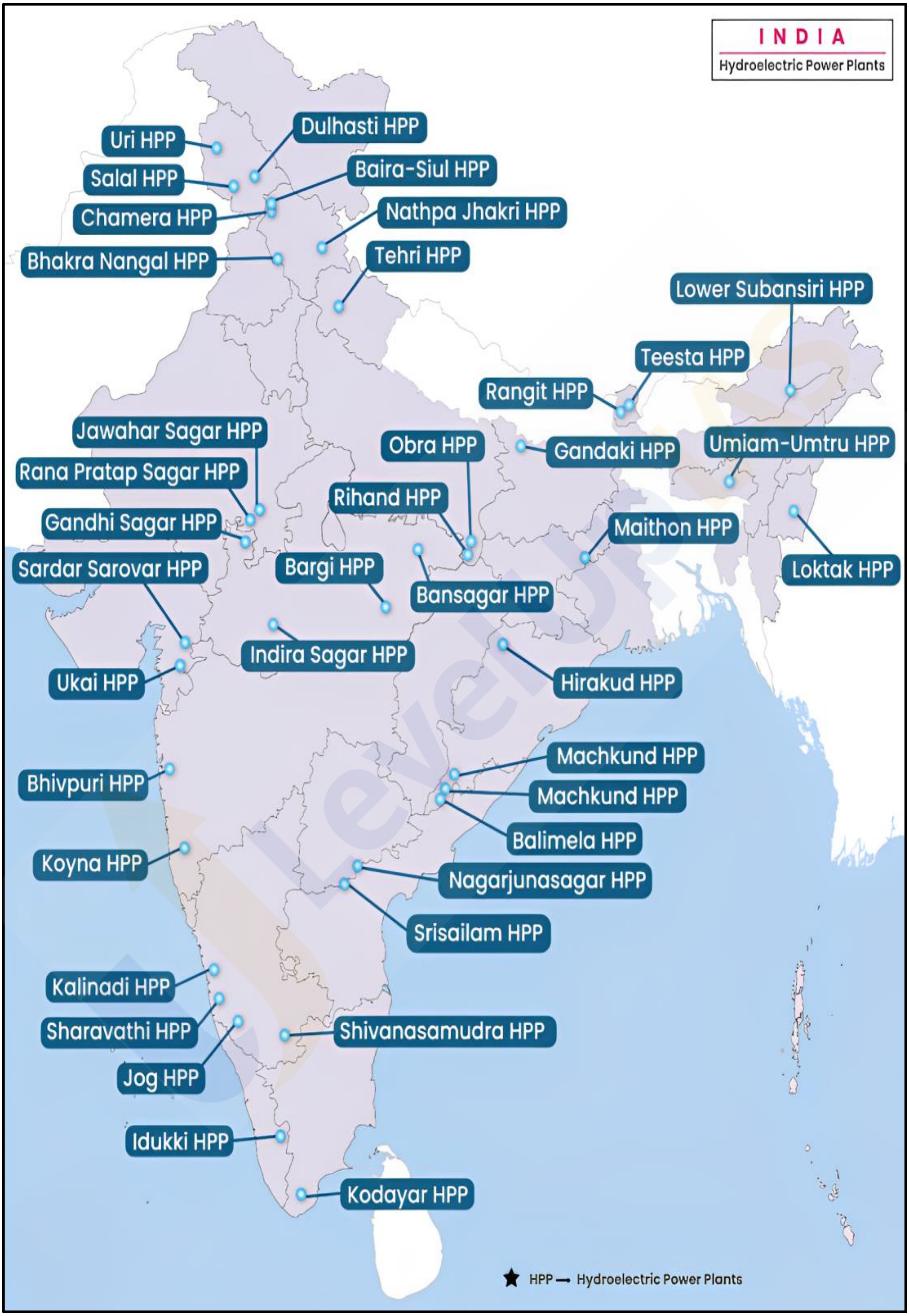
- **Commitment to Renewable Energy -** India is committed to achieving **40% of its installed capacity** from **non-fossil fuel sources by 2030**, with a target of **175 GW** of renewable energy by **2022** and **450 GW** by **2030**. Hydropower is crucial for the **grid integration of renewable energy** and **balancing energy fluctuations**.
- **Policy Reforms -** The **2008 Hydro Power Policy** encouraged **private sector participation**. In 2019, large hydropower was formally recognized as **renewable energy**, which benefits projects with a **renewable purchase obligation**. Further, the **Draft Electricity (Amendment) Bill 2020** proposes a **Hydropower Purchase Obligation (HPO)**.
- **Pumped Storage and Sustainability -** India's hydropower sector has been hailed for its role in **grid stability**. The **Teesta-V Hydropower Station** in Sikkim was recognized internationally for **sustainability** in 2019.
- **Policy Recommendations -** There is a growing call for treating hydropower as a **peaking and grid-balancing power**. Reforms should ensure **competitive tariffs, better planning, and smoother project execution** to realize the full potential of India's hydropower sector.

Important Hydroelectric Power Plants in India:

Hydroelectric Power Plant	River	State	Year of Commission	Capacity
Nagarjuna Sagar Dam	**Krishna**	Andhra Pradesh	1967	816MW
Srisailam Right Bank Power Station Project	**Krishna**	Andhra Pradesh	1982	770MW
Shivasamudram Hydro Power Project	**Kaveri**	Andhra Pradesh	1902	42MW
Karbi Langpi Hydro Electric Project (KLHEP)	**Borpani River**	Assam	2007	100MW
Kopili Hydro Electric Project	**Kopili River**	Assam	1984	275MW
Kameng Hydro Power Station	**Bichom and Tenga Rivers**	Arunachal Pradesh	2020-21	600MW
Ranganadi Hydro Power Station	**Ranganadi River**	Arunachal Pradesh	2002	405MW
Subansiri Lower Hydroelectric Project	**Subansiri**	Arunachal Pradesh	Under Construction	2000MW (Planned)
Dagmara Hydro-Electric Project	**Kosi River**	Bihar	Under Construction	130MW (Planned)
Hasdeo Bango Dam	**Hasdeo River**	Chhattisgarh	1961-62	120MW

Sardar Sarovar Project	**Narmada River**	Gujarat	2017	River Bed Power House (RBPH) – 1200 MW Canal Head Power House (CHPH) – 250MW
Bhakra Hydroelectric Project	**Satluj**	Himachal Pradesh	Bhakra Left Bank Power House – 1961 Bhakra Right Bank Power House – 1968	1325MW
Pong Hydroelectric Project	**Beas River**	Himachal Pradesh	1978	396MW
Dehar Hydroelectric Project	**Beas River**	Himachal Pradesh	1977	990MW
Nathpa Jhakri Hydro Electric Project	**Satluj River**	Himachal Pradesh	2004	1500MW
Karcham Wangtoo Hydroelectric Plant	**Satluj River**	Himachal Pradesh	2011	1091MW
Koldam Hydroelectric Plant	**Satluj River**	Himachal Pradesh	2015	800MW
Baglihar Stage- I Hydroelectric Project	**Chenab River**	Jammu & Kashmir	2008	450MW
Salal Hydro Electric Power plant	**Chenab River**	Jammu & Kashmir	1987	720MW
URI-I Uri power station	**Jhelum River**	Jammu & Kashmir	1997	480MW
The Pakal Dul Hydro Electric Project	**Marusudar**	Jammu & Kashmir	Under Construction	1000MW
Subarnarekha Hydel Power Project	**Subarnarekha River**	Jharkhand	1977	130MW
Almatti Hydroelectric Project	**Krishna River**	Karnataka	2005	290MW
Sharavathi Hydro Power Plant	**Sharavathi River**	Karnataka	1964	1035MW
Pallivasal Hydroelectric Project	**Periyar River**	Kerala	1940	37.5MW

Idukki Hydro Electric Project	**Periyar River**	Kerala	1976	780MW
Indira Sagar Hydroelectric Project	**Narmada River**	Madhya Pradesh	2004	1000MW
Bansagar Hydroelectric Project	**Sone River**	Madhya Pradesh	1991	435MW
Koyna (Pophali) Hydroelectric Project	**Kyona River**	Maharashtra	1967	1960MW
Dikhu Hydro Electric Project	**River Dikhu and River Yangnyu**	Nagaland	Under Construction	186MW
Hirakud Hydro Electric Project	**Mahanadi**	Odisha	1957	276MW
Balimela Hydro Electric Project	**Sileru River**	Odisha	1975	510MW
Anandpur Sahib Hydroelectric Project	**Satluj River**	Punjab	1985	134MW
Ranjit Sagar Dam Hydroelectric Project	**Ravi River**	Punjab	2000	600MW
Teesta-VI hydroelectric power project	**Teesta River**	Sikkim	Under Construction	500MW
Kundah Hydroelectric Project	**Bhavani river**	Tamil Nadu	1960	585MW
Kadamparai Hydroelectric Project	**Kadampari River**	Tamil Nadu	1987	400MW
Srisailam Left Bank Hydroelectric Project	**Krishna River**	Telangana	2001	900MW
Rihand Hydroelectric Project	**Rihand River**	Uttar Pradesh	1955	300MW
Tehri Hydroelectric Project	**Bhagirathi River**	Uttarakhand	2006	2400MW
Kishau Dam Hydropower Project	**Tons River**	Uttarakhand	Under Construction	660MW


INDIA
Hydroelectric Power Plants
Uri HPP
Dulhasti HPP
Salal HPP
Baira-Siul HPP
Chamera HPP
Nathpa Jhakri HPP
Bhakra Nangal HPP
Tehri HPP
Lower Subansiri HPP
Teesta HPP
Rangit HPP
Jawahar Sagar HPP
Obra HPP
Gandaki HPP
Umiam-Umtru HPP
Rana Pratap Sagar HPP
Rihand HPP
Gandhi Sagar HPP
Maithon HPP
Sardar Sarovar HPP
Bargi HPP
Loktak HPP
Bansagar HPP
Indira Sagar HPP
Ukai HPP
Hirakud HPP
Machkund HPP
Bhivpuri HPP
Machkund HPP
Balimela HPP
Koyna HPP
Nagarjunasagar HPP
Srisailam HPP
Kalinadi HPP
Sharavathi HPP
Shivanasamudra HPP
Jog HPP
Idukki HPP
Kodayar HPP
HPP — Hydroelectric Power Plants

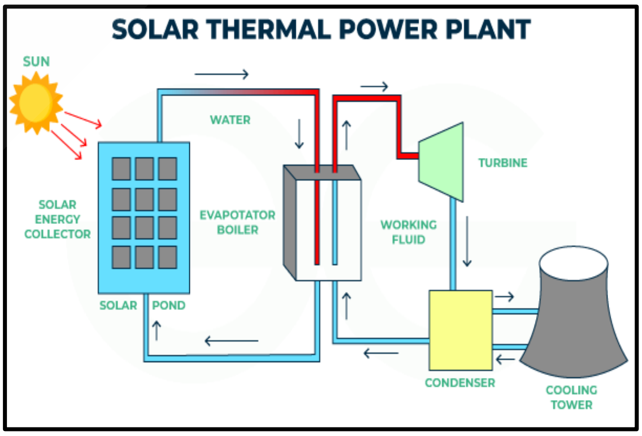
Solar Energy:

- The **Sun is the primary source of energy** on Earth, and as a **tropical country,** India is naturally endowed with an abundant supply of solar energy. Most regions of the country experience **bright sunshine throughout the year**, except during a brief monsoon period. Due to this, **solar energy plays a vital role** in India's renewable energy sector, harnessed through both **thermal** and **photovoltaic routes**. This energy can be applied in various ways, such as **cooking, water heating, drying farm produce, water pumping, lighting homes and streets, and generating power** to meet decentralized needs in rural areas, schools, hospitals, and more.
- India receives approximately **3,000 hours of sunshine annually**, which is equivalent to over **5,000 trillion kilowatt-hours (kWh)** of energy per year. This potential is **far greater than the total energy consumption of the country**. The **daily average solar energy incident** over different parts of India ranges from **4 to 7 kWh/m²**, depending on the location. As a result, solar technologies such as **solar water heaters, solar refrigeration, solar drying, street lighting, cooking, pumping, power generation**, and **photovoltaic solar cells** are becoming increasingly popular across the country.

Solar Energy Potential in India:

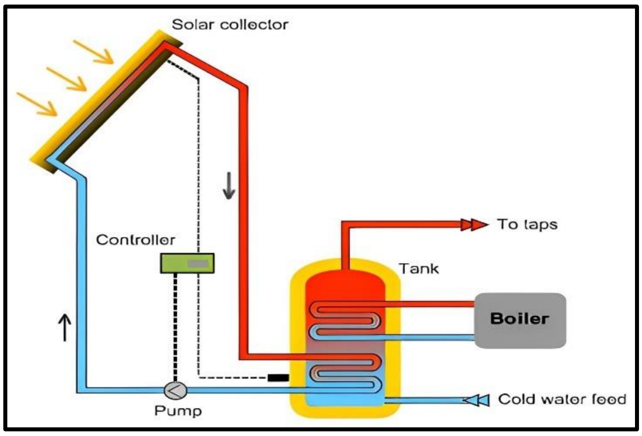
- Although solar energy can be efficiently utilized across almost all parts of the country, certain regions hold **immense potential** for solar power generation. One such region is the **Thar Desert** in Rajasthan. According to scientists, the **vast expanse of the Thar Desert** could potentially become the **largest solar power hub in the world**. The desert spans around **35,000 sq. km**, which could generate between **700 gigawatts to 2,100 gigawatts** of solar power. A significant portion of this desert area has been designated as a **'Solar Energy Enterprise Zone'**, similar to the solar energy zones in Nevada, USA.
- Other areas in India, such as the **Kathiawar Peninsula, Maharashtra, Karnataka, Andhra Pradesh, Telangana, Madhya Pradesh, West Bengal, Jharkhand, Bihar, Uttar Pradesh, Haryana, and Punjab**, also have considerable potential for **harnessing solar energy**.

Methods of Solar Energy Utilization:



Solar Thermal Power Production:

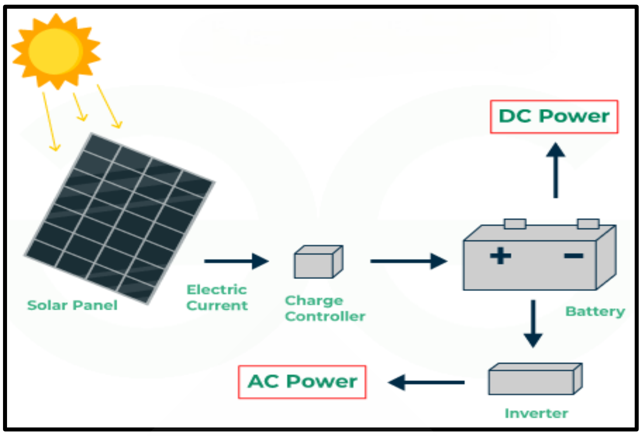
- **Solar thermal power production** involves **converting solar energy into electricity by first utilizing the sun's heat to warm a working fluid—such as water or air.** This heat energy is then used to drive a turbine, which generates mechanical energy. A conventional generator, coupled to the turbine, subsequently converts this mechanical energy into electrical energy. This method efficiently harnesses solar thermal energy for power generation.



Solar Heating Systems:

- **Solar heating systems** harness solar energy to **heat a fluid, which can be either liquid or air.** The heated fluid is then transferred directly to an interior space or to a storage system for later use. If the solar system alone cannot meet the space heating requirements, an **auxiliary or backup system is employed to provide the additional heat necessary.** This approach helps in utilizing solar energy for effective heating solutions.

Photoelectric Cells:



- **Photoelectric cells**, or photovoltaic (PV) cells, are a popular method for **converting solar energy into electricity.** These cells are primarily **silicon-based materials that absorb sunlight.** When sunlight strikes the cells, it causes electrons to move, generating an electric current. This electricity is produced in the form of **direct current (DC). PV cells** are widely used for generating renewable electricity directly from sunlight.

Advantages of Solar Energy:

- **Renewable Resource -** Solar energy is a **truly renewable source** that can be harnessed in almost every part of the world and is available daily.
- **High Energy Yield -** Solar energy systems can potentially offer **grid-free operation** if the energy produced is sufficient to meet the needs of a home or building.
- **Pollution-Free -** Solar power is **pollution-free** and does not emit greenhouse gases once the installation is complete, contributing to environmental protection.
- **Low Running Costs -** Solar energy systems have **low operating costs** and provide favorable returns on investment through grid tie-up mechanisms such as net metering.
- **Long Equipment Life -** Solar energy conversion equipment typically has a **long lifespan** and requires minimal maintenance, enhancing energy infrastructure security.

Disadvantages of Solar Energy:

- **Expensive Energy Storage -** Solar energy must often be used immediately or stored in **large, costly batteries**. These batteries, used in off-grid systems, can be charged during the day and used at night, but their expense can be significant.
- **Space Requirements -** To generate substantial amounts of electricity, a large number of solar panels are needed, which requires considerable **space**. Some roofs may not be large enough to accommodate the desired number of panels.
- **Associated Pollution -** While solar energy-related pollution is minimal compared to other energy sources, it still exists. The **transportation and installation** of solar systems can result in greenhouse gas emissions. Additionally, the **manufacturing process** of photovoltaic systems involves some toxic materials and hazardous products, which can indirectly impact the environment.

Scenario of Solar Energy in India:

- India, positioned in the **tropical belt**, is endowed with the potential to harness solar energy, receiving peak solar radiation for **approximately 300 days** a year. This translates to around **2,300 to 3,000 hours of sunshine**, amounting to over **5,000 trillion kWh** of solar energy potential.
- The solar power sector in India is experiencing rapid growth as a critical component of the country's **renewable energy** landscape. As of **31 August 2021**, India's **solar installed capacity** reached **44.3 GW**. To facilitate the expansion of solar projects, the country has established nearly **42 solar parks**, providing land to solar plant promoters.

Commitment to Renewable Energy Goals:

- As part of its **Intended Nationally Determined Contributions (INDCs)**, India aims to achieve **100 GW** of solar power out of a total of **175 GW** of renewable energy by **2022**. This sector also holds significant potential for job creation, with an estimated **4,000 direct and indirect jobs** generated per **1 GW** of solar manufacturing capacity. Additionally, roles in **solar deployment, operation, and maintenance** contribute to recurring employment opportunities.
- **India is projected to account** for **8%** of **global solar capacity** by **2035**. With a future potential capacity of **363 GW**, the nation is positioned to become a global leader in capitalizing on energy sector advantages. Given the country's challenges in meeting energy demand, solar energy can play a pivotal role in ensuring **energy security**.
- The ongoing debate surrounding **global warming** and **climate change** emphasizes the need to transition from fossil fuels to **clean and green energy**. With its **pollution-free** nature and **virtually inexhaustible supply**, solar energy stands out as an attractive energy resource.

Government Initiatives:

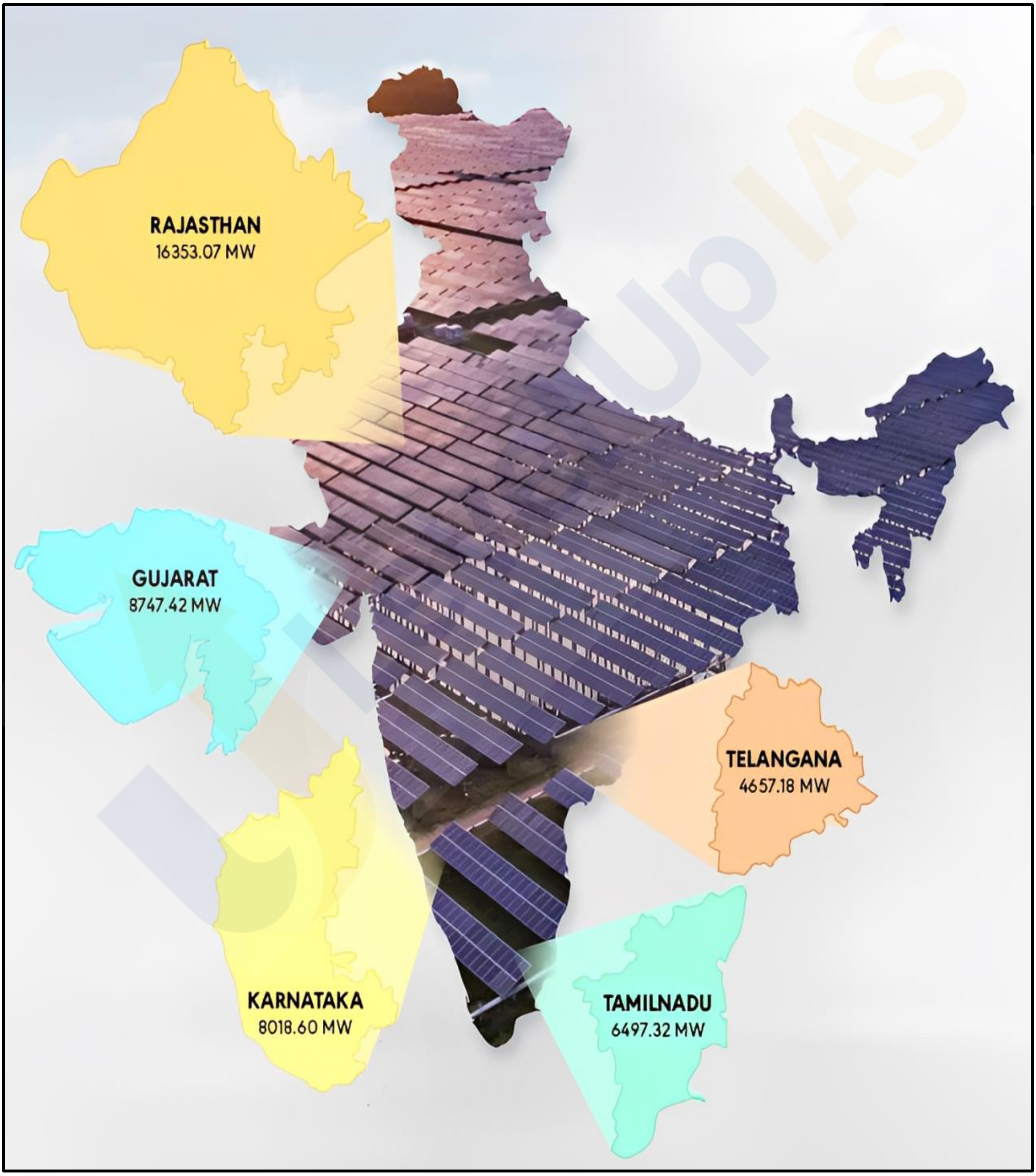
- The **Ministry of New and Renewable Energy (MNRE)** serves as the nodal agency addressing India's renewable energy challenges. Key initiatives include:
    - **National Solar Mission -** A significant initiative aimed at promoting sustainable growth while addressing energy security concerns.
    - **Indian Renewable Energy Development Agency (IREDA) -** This non-banking financial institution provides term loans for renewable energy and energy efficiency projects.
    - **National Institute of Solar Energy -** An autonomous institution under MNRE, focused on research and development.
    - **Establishment of Solar Parks -** Initiatives to create ultra-major solar power projects and enhance grid connectivity infrastructure.
    - **Canal Bank and Canal Tank Solar Infrastructure -** Promotion of solar installations on canal banks and tanks.
    - **Sustainable Rooftop Implementation of Solar Transfiguration of India (SRISTI) -** A scheme aimed at promoting rooftop solar power projects.
    - **Suryamitra Programme -** A training program designed to prepare a qualified workforce for the solar sector.
    - **Renewable Purchase Obligation -** A mandate for large energy consumers to procure a certain percentage of their energy from renewable sources.
    - **National Green Energy Programme and Green Energy Corridor -** Initiatives to support the development and integration of renewable energy.
- India's commitments at the **Paris Climate Deal** include reducing the **emissions intensity** of its GDP by **33 to 35%** from **2005 levels** by **2030**. Furthermore, the country aims to achieve **40%** of its cumulative electric power installed capacity from **non-fossil fuel-based** energy resources by **2030**. This goal will be supported by technology transfers and low-cost international financing, including resources from the **Green Climate Fund**.

International Solar Alliance (ISA):

- India has initiated the **International Solar Alliance**, comprising over **122 countries**, primarily those situated between the **Tropic of Cancer** and the **Tropic of Capricorn**. The ISA aims to promote solar energy globally and mobilize over **US $1,000 billion** in investments by **2030** for the extensive deployment of solar energy technologies.

Top Solar States in India:

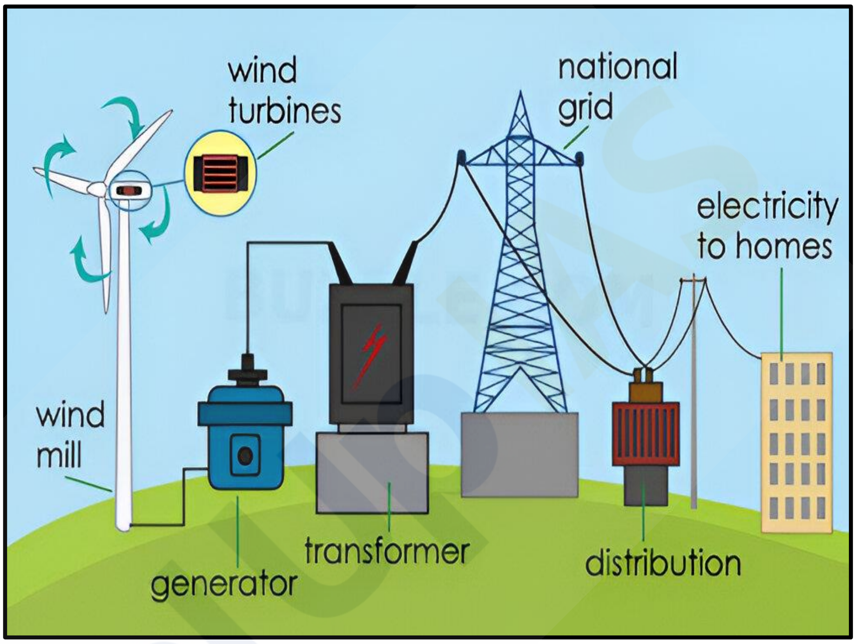
Rank	State	Solar Capacity	Prominent Solar Plant
1	Rajasthan	23 GW	Bhadla Solar Park
2	Gujarat	10.13 GW	Charanka Solar Park
3	Karnataka	9.05 GW	Pavagada Solar Park
4	Tamil Nadu	8.1 GW	Kamuthi Solar Power Project
5	Maharashtra	4.8 GW	Sakri Solar Power Plant



Wind Energy:

- **Wind energy** is defined as the **kinetic energy** associated with the **movement of atmospheric air.** Wind turbines play a crucial role in harnessing this energy, transforming it into **mechanical power** and subsequently converting it into **electric power** for electricity generation.
- Wind energy is recognized as a significant source of **renewable energy** in India. Notably, many of the largest operational onshore wind farms are located in the **United States**, **India**, and **China**. Together, five countries—**Germany**, **USA**, **Denmark**, **Spain**, and **India**—account for **80%** of the world's installed wind energy capacity.

How Wind Energy Works:



- Wind energy conversion devices, primarily **wind turbines**, are employed to convert wind energy into mechanical energy. The basic components of a wind turbine include:
  - **Blades/Vanes -** These are the sails or wings that capture the wind.
  - **Central Axis -** The blades are attached to a central hub that allows them to rotate.
- When the wind blows against the blades, they **rotate around the central axis.** This **rotational motion** can be utilized to perform various types of useful work. By connecting the wind turbine to an **electric generator**, wind energy is effectively converted into **electric energy**.
- One of the advantages of wind energy is its **high efficiency** when compared to traditional heat engines, making it a favourable option for sustainable energy production.

Wind Farm/Park:

- A **wind farm**, also known as a **wind park**, **wind power station**, or **wind power plant**, refers to a collection of wind turbines situated in the same area, working together to produce electricity. Wind farms can vary significantly in size, ranging from a few turbines to several hundred, often covering extensive land areas.

Types of Wind Farms:

- **Onshore Wind Farms -** These wind farms harness the energy of moving air through turbines **located on land.** Onshore wind energy is a **widely adopted method** for generating electricity.
- **Offshore Wind Farms - Located out at sea or in freshwater,** offshore wind turbines can either be:
  - **Fixed-Foundation Turbines -** Installed in shallow waters.
  - **Floating Wind Turbines -** Built in deeper waters, where the turbine's foundation is anchored to the seabed. Floating wind farms are still in the early stages of development.
- **Offshore wind farms** must comply with specific regulations, including being located at least **200 nautical miles** from the shore and in waters that are **50 feet deep** or more. The **electricity generated** by offshore wind turbines is transmitted back to the shore through cables buried in the ocean floor.

Offshore Wind Energy:

Advantages:

- **Larger and Taller Turbines -** Offshore windmills can be constructed larger and taller than their onshore counterparts, enabling them to capture **more energy**.
- **Less Intrusiveness -** Located far out at sea, offshore wind farms are less intrusive to neighboring countries, allowing for the development of **larger farms** per square mile.
- **Higher Wind Speeds -** Offshore sites generally experience **higher wind speeds**, leading to increased energy generation potential.
- **Unobstructed Wind Flow -** Offshore wind farms are free from physical obstructions such as hills or buildings that could impede wind flow, maximizing energy collection.

Disadvantages:

- **High Costs -** One of the most significant drawbacks of offshore wind farms is their **high construction and maintenance costs**. The challenging locations can also make them vulnerable to damage from severe storms or hurricanes, leading to **expensive repairs**.
- **Environmental Impact -** The effects of offshore wind farms on **marine life and birds** are not yet fully understood, raising environmental concerns.
- **Proximity Issues -** Offshore wind farms located closer to coastlines (generally within **26 miles**) may face opposition from residents due to their potential impact on **property values and tourism**.

Onshore Wind Energy:

Advantages:

- **Lower Costs -** Onshore wind farms are relatively inexpensive to develop, enabling the establishment of **massive wind turbine farms**.
- **Reduced Voltage Drop -** The shorter distance from the wind turbines to consumers minimizes **voltage drop-off** in the cabling, enhancing efficiency.
- **Rapid Installation -** Wind turbines can be installed much more quickly compared to other energy sources, such as nuclear power stations, which can take over twenty years. A wind turbine can often be constructed in a matter of **months**.

Disadvantages:

- **Aesthetic Concerns -** Many people consider onshore wind farms to be an **eyesore**, detracting from the landscape.
- **Variable Energy Production -** Onshore wind farms may not produce energy consistently throughout the year due to **inconsistent wind speeds** or physical barriers such as buildings and hills.
- **Noise Pollution -** The sound produced by wind turbines is often compared to that of a **lawnmower**, leading to **noise pollution** that can affect nearby communities.

Scenario of Wind Energy in India:

- India's **wind energy sector** has made significant strides, primarily driven by its **indigenous wind power industry**. The sector has developed a robust ecosystem, demonstrating strong project operation capabilities and a manufacturing base capable of producing approximately **10,000 MW annually**.
- As of **31st March 2021**, India ranks **fourth** globally in wind installed capacity, boasting a total capacity of **39.25 GW**. During the fiscal year **2020-21**, India generated around **60.149 billion units** of electricity from wind energy.
- The sector has witnessed a **compound annual growth rate (CAGR)** of **11.39%** for wind generation and **8.78%** for installed capacity between **2010 and 2020**. According to the **Ministry of New and Renewable Energy (MNRE)**, India has the potential to generate **127 GW of offshore wind energy** along its **7,600 km coastline**. The **National Institute of Wind Energy (NIWE)** estimates a total wind energy potential of **302 GW** at a hub height of **100 meters**.
- The majority of commercially exploitable wind resources (over **95%**) are concentrated in seven states: **Andhra Pradesh, Gujarat, Karnataka, Madhya Pradesh, Maharashtra, Rajasthan,** and **Tamil Nadu**. The MNRE has set ambitious targets of installing **5 GW of offshore wind energy by 2022** and **30 GW by 2030**.

Government Initiatives:

National Wind-Solar Hybrid Policy:

- The **National Wind-Solar Hybrid Policy, 2018** aims to promote large grid-connected wind-solar photovoltaic (PV) hybrid systems. Its goals include:
  - **Optimal and efficient utilization** of wind and solar resources
  - Enhanced **transmission infrastructure**
  - Effective **land use** strategies

National Offshore Wind Energy Policy:

- Introduced in **October 2015**, the **National Offshore Wind Energy Policy** is designed to develop offshore wind energy within the **Indian Exclusive Economic Zone (EEZ)**. The policy **aims to:**
  - **Explore and promote** offshore wind farm deployment, including public-private partnerships.
  - **Reduce carbon emissions** by transitioning to offshore wind energy.
  - **Attract investments** in energy infrastructure to enhance energy security.
  - **Encourage spatial planning** and management of offshore renewable resources.
  - **Indigenize** offshore wind energy technology.
  - **Promote research and development** in the offshore wind sector.
  - **Create a skilled workforce** and employment opportunities in the offshore wind energy sector.
  - Facilitate the development and maintenance of **coastal infrastructure** to support offshore projects.

Challenges in the Wind Energy Sector in India:

- **Recent Decline in Growth -** In **2016-17**, the sector added approximately **5.5 GW**, which dropped to **2 GW** in **2017-18**.
- **Diminishing Incentives -** Initial growth was driven by generous incentives, including generation-based incentives and accelerated depreciation, which have been gradually reduced.
- **Investment Competition -** The cost of solar energy has significantly decreased, with solar power priced at **Rs 2.23** per unit compared to **Rs 4.50** for wind energy, making solar more appealing to investors.

- **Transitional Policies -** Wind energy policies are still evolving, and auction frameworks introduced in **December 2017** impose **tariff ceilings**, complicating tariff achievement due to region-specific characteristics.
- **Distribution Company Risks -** General challenges related to distribution companies (discoms), such as power generation curtailments and delayed payments, pose risks for investors.
- **Specific Challenges in Offshore Wind Energy**:
  - **Data Collection Requirements -** Extensive metocean and geological data must be collected prior to project initiation, as seen in Germany, where data collection took eight years before project commencement.
  - **Infrastructure Investments -** Significant investments are needed for developing support infrastructure.
  - **Higher Installation Costs**: Offshore wind turbines require stronger structures than their onshore counterparts, leading to increased installation costs due to a lack of local substructure manufacturers and trained personnel.
  - **Maintenance Costs -** Offshore turbines are subject to harsh environmental conditions, necessitating costly maintenance efforts.
- Currently, **Europe leads in offshore wind energy,** with tariffs comparable to India's onshore rates, supported by longstanding incentives absent in India. There are concerns about attracting investment for offshore projects amid the current low costs of solar energy and a relatively undeveloped onshore wind sector.

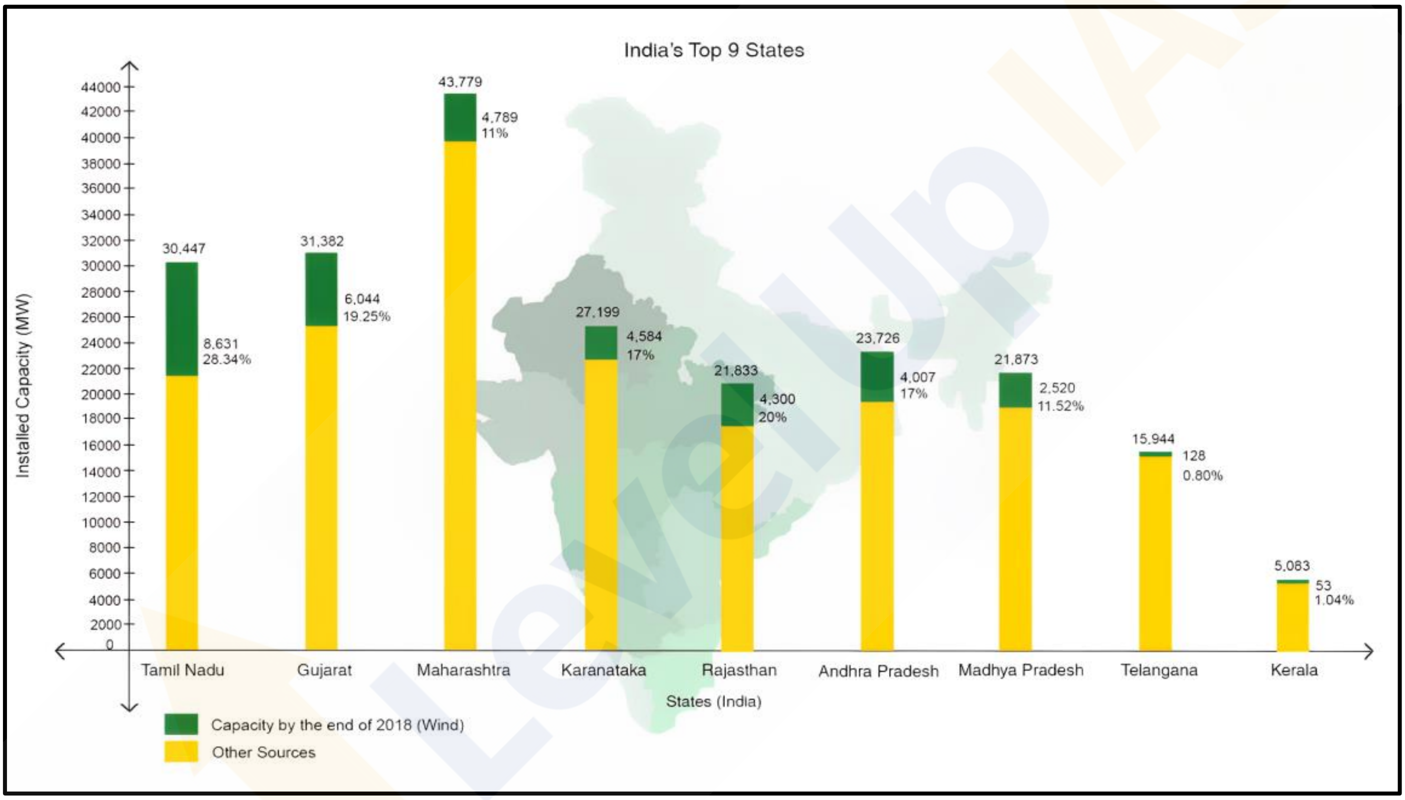
Way Forward:

- **Renewable Purchase Obligation -** Power distribution companies, open access consumers, and captive users should be mandated to purchase a certain percentage of their total electricity consumption as clean energy through a **renewable purchase obligation**.
- **Lower Taxes -** Although electricity and power sales are exempt from GST in India, wind power generation companies cannot claim input tax credits for GST paid on project setup. Addressing this could stimulate investment.
- **Feed-in Tariff (FiT) -** Adopting **feed-in tariff regulations** can make offshore wind power procurement mandatory for discoms. FiT can be tailored for each offshore project to support early development until the sector becomes economically viable.
- **Deemed Generation Provision -** Implementing a **deemed generation provision** for offshore wind projects can protect against power generation curtailment due to limitations in State Load Dispatch Centres' (SLDCs) ability to absorb excess power.
- **Development of Sub-Sea Substations -** The development of **sub-sea substations** by the Power Grid Corporation of India Ltd can mitigate risks for offshore wind farm developers and improve the feasibility of offshore projects.

Top States in India (Installed Wind Power Capacity):

- **Tamil Nadu -** Tamil Nadu tops the list of states with the **largest installed wind power generation capacity** in the country. Share of wind power in electricity generation was around 28% in 2018. Total wind capacity at the end of 2018 stood at 8,631 MW while its total installed electricity generation capacity stood at **30,447 MW at the end of 2018.**
- **Gujarat -** Gujarat houses the **second-largest installed wind power generation** capacity in the country. Share of wind power in electricity generation was around **19%** in 2018.
- **Maharashtra -** Maharashtra houses the **third-largest installed wind power generation capacity** in the country.
- **Karnataka -** Karnataka houses the **fourth-largest installed wind power generation capacity** in the country.

- **Rajasthan -** Rajasthan houses the **fifth-largest installed wind power generation capacity** in the country. Wind contributes around 20% of total electricity generated in the state.
- **Andhra Pradesh -** Andhra Pradesh houses the **sixth-largest installed wind power generation capacity** in the country. The state's wind capacity at the end of 2018 stood at 4,007 Mw while its total installed electricity generation capacity stood at 23,726 Mw at the end of 2018, with
- **Madhya Pradesh -** Madhya Pradesh houses the **seventh largest installed wind power generation capacity** in the country. The state's wind capacity at the end of 2018 stood at 2,520 Mw while its total installed electricity generation capacity stood at 21,873 Mw at the end of 2018, with the wind sector's share at 11.52 per cent.
- **Telangana -** Telangana's installed wind capacity at the end of 2018 stood at 128 Mw while its total installed power generation capacity stood at 15,944 Mw at the end of 2018, with the wind sector's share at 0.80 per cent.
- **Kerala -** Kerala's installed wind capacity at the end of 2018 stood at 53 Mw while its total installed power generation capacity stood at 5,083 Mw at the end of 2018, with the wind sector's share at 1.04 per cent.



Geothermal Energy:

- **Geothermal energy** refers to the **natural heat generated from the Earth's interior,** which can be harnessed for **electricity generation** and **heating buildings**. This energy is derived from the **magma**, a layer of hot, molten rock located beneath the Earth's crust. The heat produced in this layer primarily results from the decay of naturally radioactive materials such as **uranium** and **potassium**. Remarkably, the heat available within **10,000 meters** (about **33,000 feet**) of the Earth's surface holds approximately **50,000 times more energy** than all the oil and natural gas resources combined.

Categories of Geothermal Resources

- **Geopressurized zones**
- **Hot-rock zones**
- **Hydrothermal convection zones**

Among these, only **geopressurized zones** are currently exploited on a commercial scale. Natural examples of Geothermal Energy include **Geysers, Lava fountains and Hot springs etc.**

Advantages of Geothermal Energy:

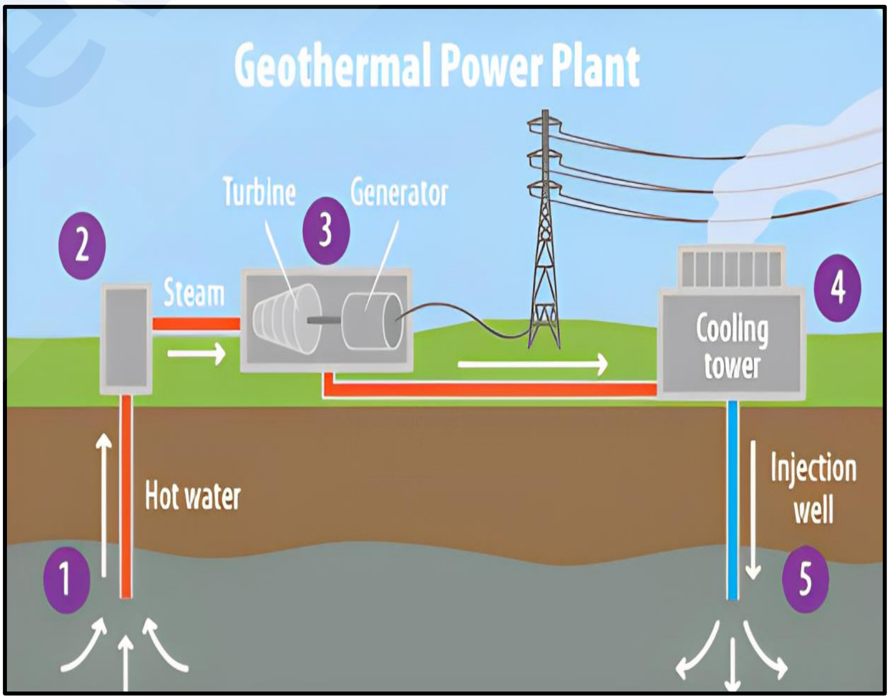
- **Renewable Resource - Geothermal energy** is considered a **renewable resource**. It is estimated that the heat radiating from the Earth's core can satisfy human energy demands for an indefinite period.
- **Ease of Exploitation -** Historically, geothermal energy has been accessible for various uses, such as **bathing, home heating**, and **cooking**. Today, it is increasingly utilized for the **direct heating** of homes and offices, making it an affordable energy option.
- **Reduced CO2 Emissions -** Another significant advantage is that geothermal energy produces **less CO2** compared to fossil fuels. Its use contributes to **pollution reduction** and supports efforts against **global warming**.
- **High Net Energy Yield -** With the increasing demand for energy, geothermal systems allow for additional units with **modular designs** to be installed easily. The cost of electricity production from geothermal sources is nearly **competitive with conventional energy sources**.

Disadvantages of Geothermal Energy:

- **Limited Availability -** One of the primary drawbacks of geothermal energy is that it is **not available everywhere**. Geothermal hotspots are often located in remote regions far from energy-demanding areas.
- **H2S Pollution -** Geothermal energy extraction can release significant amounts of **hydrogen sulfide (H2S)**, a toxic gas often associated with a smell reminiscent of rotten eggs. High concentrations of H2S can be lethal.
- **Water Pollution -** The process of harnessing geothermal energy can also lead to **water pollution**, akin to the impacts of mining operations.

How Geothermal Energy is Captured?

- Geothermal systems are typically found in regions with a normal or slightly elevated **geothermal gradient**, defined as the **temperature increase with depth in the Earth's crust.** This average gradient is about **2.5-3 °C per 100 meters**. The most common method for capturing geothermal energy involves tapping into naturally occurring **hydrothermal convection** systems. Here, cooler water seeps into the Earth, gets heated, and rises back to the surface. This heated water can then be harnessed to drive **electric generators**.



Challenges of Geothermal Energy:

- **Resource and Location Limitations -** While geothermal energy exists beneath the Earth's surface, it is not uniformly accessible. Only a fraction of land is situated above suitable geothermal reservoirs, limiting the establishment of geothermal power plants. Additionally, many promising geothermal sites are in tectonically active areas, posing risks associated with **earthquakes** and **volcanic activity**, which deter investment in large-scale facilities.
- **Infrastructure Constraints -** A significant barrier to the widespread use of geothermal energy, particularly in the United States, is the lack of appropriate **infrastructure**. Geothermal energy typically serves as a **baseline power** source for electrical grids, complicating its integration. The **cost of drilling wells** and constructing power plants is prohibitively high, and training personnel to operate these facilities can be resource-intensive. Furthermore, the energy extracted cannot be easily transported to areas with higher demand; it must be utilized immediately.
- **High Initial Costs -** The financial investment required to tap geothermal energy is substantial, with costs ranging from **$2 million to $7 million** for a **1-megawatt** capacity plant. However, these high upfront costs can be justified as a long-term investment.
- **Limited Resource Capacity -** Despite its classification as renewable, geothermal energy is not **unlimited**. Each well has a finite capacity for water extraction, and without proper reinjection of the used water, pressure may decline, resulting in reduced energy output and potential geological issues, such as **sinkholes**.
- **Location Restrictions -** A major limitation of geothermal energy is its **location specificity**. Power plants must be established in areas where geothermal resources are available, which can exclude many regions from accessing this energy source.
- **Transmission Challenges -** Geothermal plants need to be situated near geothermal reservoirs, as transporting steam or hot water over distances greater than **two miles** is impractical. This geographic limitation, combined with the fact that optimal geothermal sites are often in rural areas, can hinder the ability to connect these facilities to power grids.
- **Environmental Concerns -** Although geothermal energy generally has a lower carbon footprint, the extraction process can release greenhouse gases stored beneath the Earth's surface. Although the emissions are considerably lower than those associated with fossil fuels, the rate of gas release may increase near geothermal facilities.
- **Earthquake Risks -** There is also a risk of inducing **earthquakes** through geothermal energy extraction, particularly with enhanced geothermal systems that inject water into the Earth's crust. Although such activities are usually conducted away from populated areas, the implications of induced seismic activity must still be considered.
- **Sustainability Issues -** For geothermal energy to remain sustainable, fluid must be reintroduced into underground reservoirs faster than it is depleted. Proper management is essential to maintain the sustainability of geothermal resources.

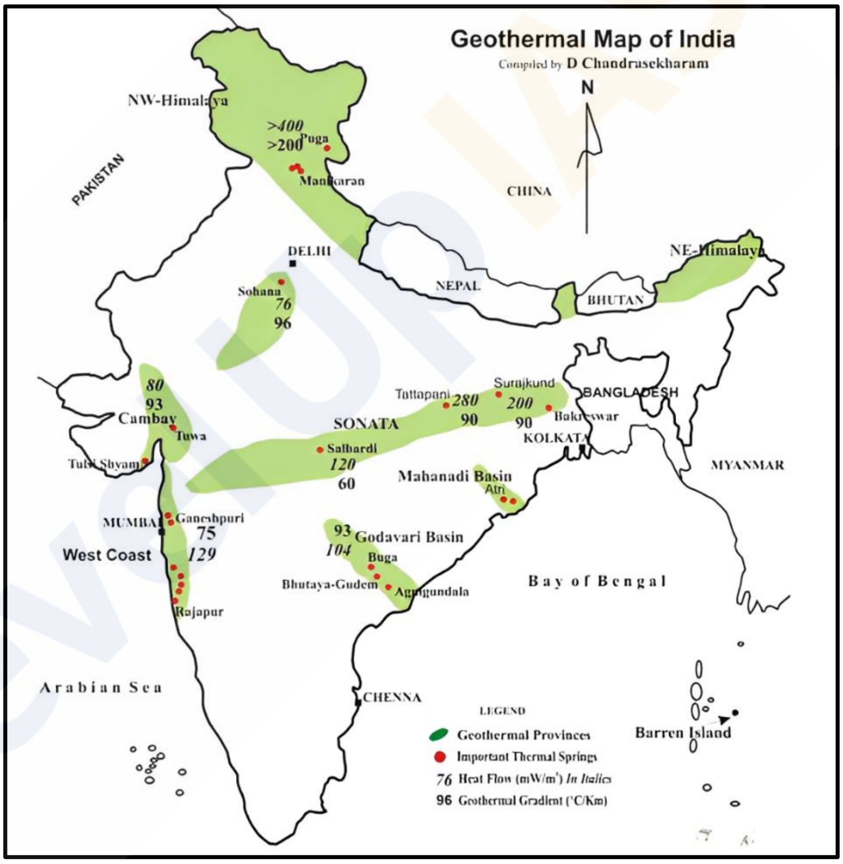
Geothermal Energy in India:

- Geothermal energy is a promising renewable energy source in India, with exploration and research dating back to **1970**. The **Geological Survey of India (GSI)** has identified approximately **350 geothermal energy locations** throughout the country, highlighting the potential for harnessing this energy.

Geothermal Provinces:

- India is home to **seven geothermal provinces**, each with unique characteristics and potential for geothermal energy generation:
    - **Himalayas**
    - **Sohana**
    - **West Coast**
    - **Cambay (Gujarat)**
    - **Godavari**
    - **Mahanadi**
    - **Son-Narmada-Tapi (SONATA)**

Geothermal Energy Potential in India:



- The **GSI** has mapped the geothermal resources in India, estimating a potential of around **10 gigawatts (GW)** for geothermal power generation, as per the **Ministry of New and Renewable Energy (MNRE)**. This estimate underscores the significant opportunity for expanding renewable energy capacity through geothermal sources.
- Satellites, particularly the **IRS-1**, have played a crucial role in identifying geothermal areas through **infrared photographs**, facilitating better exploration and development efforts.

Geothermal Power Plants:

- In **2013**, the **Chhattisgarh government** announced plans to establish a geothermal power plant in **Tattapani**, located in the **Balrampur district**.
- In **2021**, an agreement was signed to initiate the **first geothermal power project in Ladakh**, further enhancing India's geothermal capabilities.

Government Initiatives:

- A capital subsidy of **up to 30%** for industrial projects.
- The establishment of the first geothermal power plant in Chhattisgarh through a joint effort between **NTPC** and the **Chhattisgarh Renewable Energy Development Agency (CREDA)**, located in the **Tattapani geothermal field** within the SONATA geothermal province.
- The **MNRE** provides significant incentives and subsidies for **Research, Design, Development, and Demonstration (RDD&D)** to enhance geothermal energy harnessing technologies.
- A target has been set by the MNRE to generate **up to 1000 MW** of geothermal energy by **2022**.

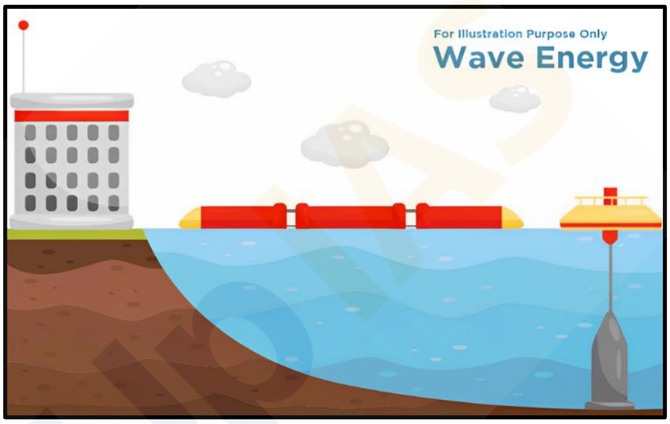
Ocean Energy:

- **Ocean energy** encompasses all forms of **renewable energy** derived from the sea, providing a promising solution to meet the increasing global energy demands. The **Ministry of New and Renewable Energy** in India has recognized ocean energy as a viable form of renewable energy. There are three primary types of ocean energy technologies: **wave energy**, **tidal energy**, and **ocean thermal energy**.

Types of Ocean Energy:

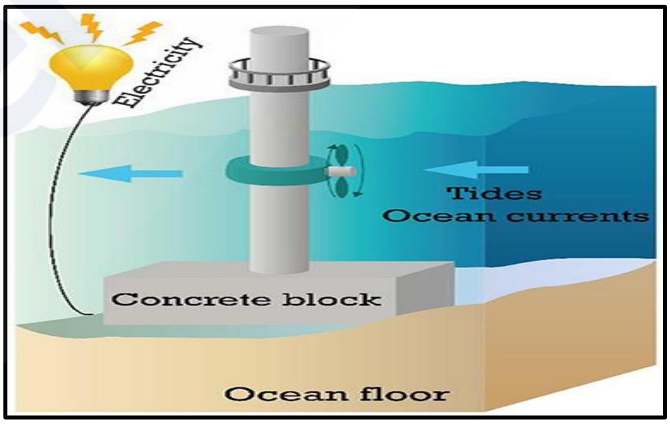
Wave Energy:

- **Wave energy** is generated by **converting the kinetic and potential energy present in ocean waves, or swells, into electricity.** Various technologies are being developed and tested to effectively harness this energy. The **first wave energy project** in India, with a capacity of **150 MW**, has been established at **Vizhinjam** near **Trivandrum**. However, the development of wave energy remains limited globally, and the potential of ocean waves has not been fully exploited for human use, primarily serving as power supplies for buoys and navigational aids. Wave power is created through the **up-and-down motion** of floating devices positioned on the ocean's surface, which capture the natural movements of ocean currents and swells to generate electricity.



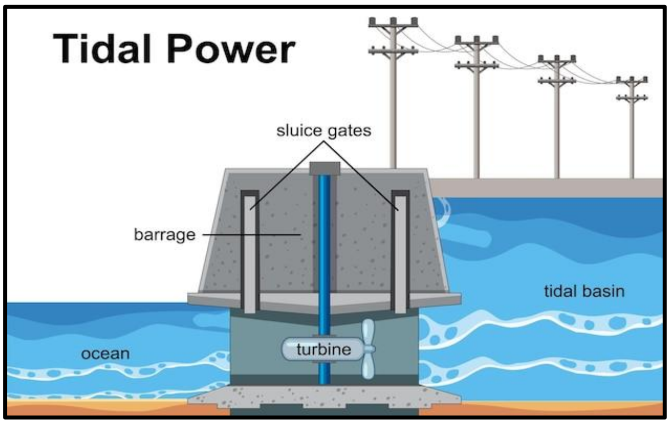
Current Energy:

- Similar to wind energy, **current energy** is generated from underwater marine currents. Large turbines, akin to propellers, are tethered to the seabed and moved by these currents to produce electricity. According to the **Intergovernmental Panel on Climate Change (IPCC)**, the potential for project-scale growth is significant as technologies continue to harness lower-velocity currents in open oceans.



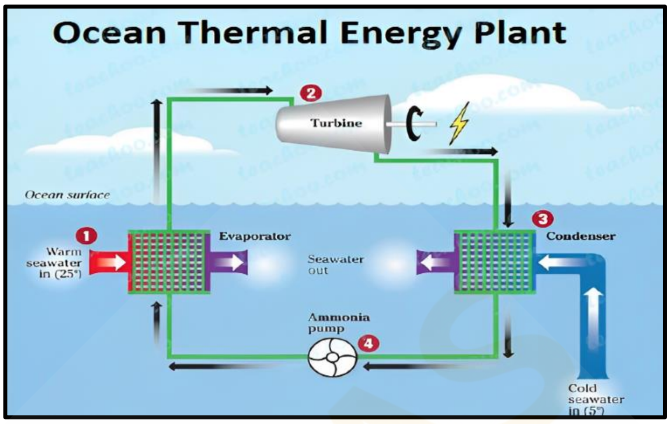
Tidal Energy:

- **Tidal energy** harnesses the periodic rise and fall of ocean water levels, driven by the **gravitational effects** of the sun and moon. This gravitational force results in tides that can be utilized to generate electricity. Like conventional **hydroelectric dams**, power plants for tidal energy are typically constructed on river estuaries, where they can hold back vast amounts of tidal water, releasing it to generate power. India is estimated to have a **tidal energy potential** of **9,000 MW**, particularly in areas like the **Gulf of Kutch** in Gujarat, the **Gulf of Cambay**, and the **Sundarbans** in West Bengal, where the tidal height is sufficient for constructing economically viable power plants.



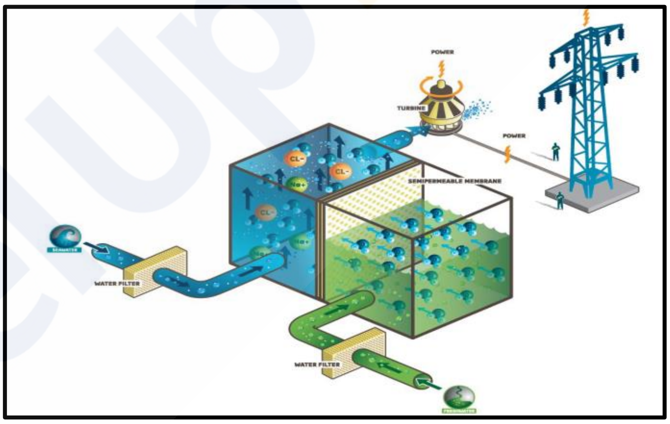
- To extract energy from tides, reservoirs or basins are created behind barrages, allowing tidal waters to pass through turbines to generate electricity. A significant tidal power project, with an estimated cost of **Rs. 5000 crores**, is proposed for the **Gulf of Kutch**.

Ocean Thermal Energy:



- The oceans serve as vast **heat reservoirs**, covering approximately **70%** of the Earth's surface. The temperature difference between warm surface waters and cold deeper layers can be utilized to generate steam and produce power. On average, the tropical seas absorb solar radiation equivalent to the heat content of **245 billion barrels of oil**, making this energy source particularly valuable.
- The process of harnessing this energy is known as **Ocean Thermal Energy Conversion (OTEC)**, which utilizes temperature differences between the ocean surface and depths of about **1000 meters** to operate a heat engine that generates electricity.

Osmotic Energy:



- **Osmotic energy** is another innovative approach to harnessing energy from the ocean. This technique generates energy from the movement of water across a **membrane** between saltwater and freshwater reservoirs, utilizing the **salinity gradient**. This method, also referred to as **salinity gradient energy**, highlights the diverse potential of ocean energy technologies in contributing to sustainable energy solutions.

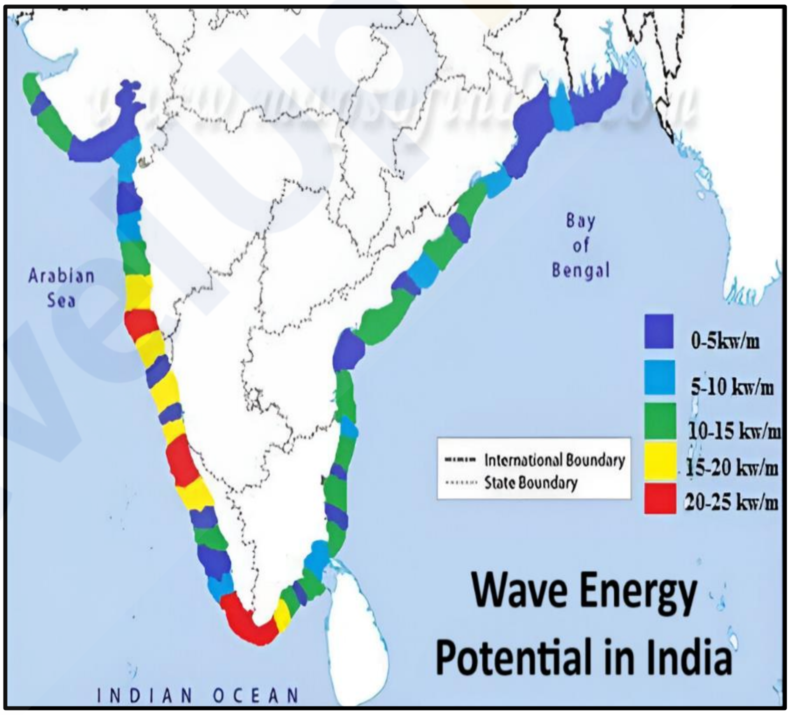
Features of Ocean Energy:

- **Predictable and Reliable - Ocean energy sources**, such as tidal and wave energy, are known for their **predictability and reliability**. Unlike wind energy, which can be intermittent, ocean currents provide a continuous flow of energy, ensuring a **consistent supply** for future use.
- **Global Presence - Tidal streams and ocean currents** are accessible across the globe, making them a **widely available resource**. This universal presence allows for the potential harnessing of ocean energy in various regions worldwide.
- **Energy-Rich -** The **kinetic energy** of moving water is significantly greater than that of moving air, as water is more than **800 times denser** than air. This density increase enables ocean energy to offer **substantial amounts of energy**, enhancing its viability as a sustainable energy source.
- **Unlimited Usage Area -** As land becomes a **scarce resource** in many regions, onshore energy solutions face competition and limitations. In contrast, **ocean energy** is derived from vast and deep oceans, eliminating this competition and providing an almost **unlimited usage area** for energy generation.

Challenges in Harnessing Ocean Energy:

- **Limited Deployment -** Currently, the **deployment of ocean energy technologies** in India is limited, with many systems not reaching their full potential. The existing technologies often suffer from **underutilization**, hindering overall energy production.
- **Research and Development Gaps -** There is a notable lack of extensive research on ocean energy technologies, with many initiatives still in the **early stages of research and development (R&D)**, demonstration, and commercialization. This stagnation restricts advancements and implementation.
- **Environmental and Commercial Risks -** The **marine environment** presents various uncertainties and risks, including:
    - **Corrosion of materials** due to seawater salinity.
    - **Challenges in offshore maintenance**.
    - **Environmental impacts** on coastal landscapes and marine ecosystems.
    - **Competition from other marine activities**, such as fishing, which may interfere with energy production.

Potential of Ocean Energy in India:



- **Tidal Energy** The **total identified potential of tidal energy** in India is approximately **12,455 MW**. Notable locations for harnessing this energy include the **Khambat and Kutch regions**, as well as various **large backwaters**, where barrage technology can be effectively utilized.
- **Wave Energy -** The **total theoretical potential** of wave energy is estimated to be around **40,000 MW**. However, it is important to note that this energy is generally less intensive than what is available in **higher latitudes**.
- **Ocean Thermal Energy Conversion (OTEC) - OTEC** holds a theoretical potential of **180,000 MW** in India, contingent on the evolution of suitable technologies. This vast potential could significantly contribute to the energy landscape.

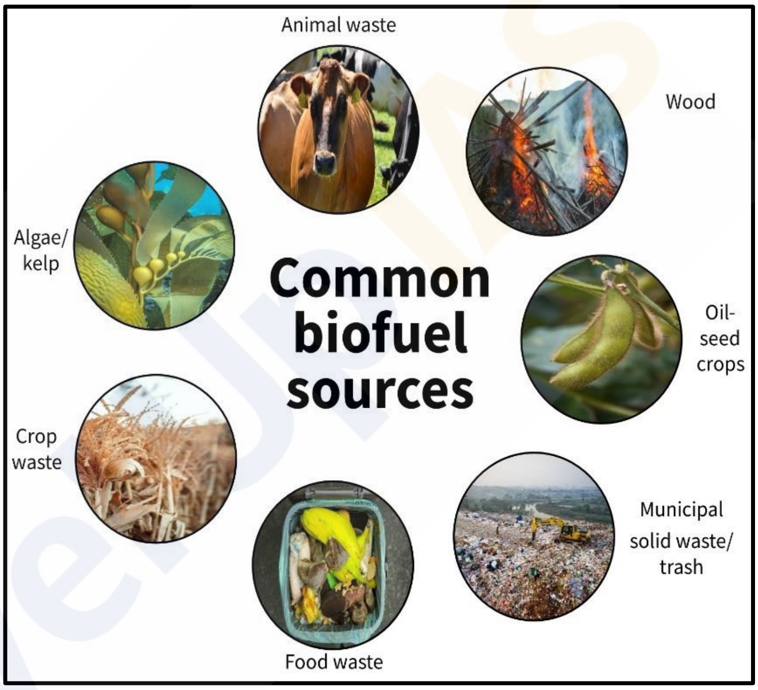
Way Forward:

- **Utilization of India's Coastline -** Given India's extensive coastline and numerous estuaries and gulfs, there is a tremendous opportunity to fully harness **ocean energy**.
- **Large-Scale Electricity Generation - Tidal streams and ocean currents** represent vast resources that can be employed for large-scale electricity generation with minimal **environmental interactions**.
- **Collaborative R&D Initiatives -** While the **National Institute of Ocean Technology in Chennai** oversees basic R&D under the **Ministry of Earth Sciences**, increased collaboration with other prominent institutions could expedite the development of ocean energy technologies and enhance their effectiveness.

Biomass Energy:

- **Biomass energy** is a form of renewable energy derived from organic materials sourced from plants and animals. This energy source continues to play a significant role in various countries, particularly in **developing nations**, where it is primarily utilized for **cooking and heating**. In contrast, many **developed countries** are increasingly adopting biomass fuels for **transportation** and **electricity generation** as a strategy to mitigate **carbon dioxide emissions** associated with fossil fuel consumption.
- Biomass contains **stored chemical energy** captured from the sun, which plants convert through the process of **photosynthesis**. This energy can be harnessed in multiple ways: it can be **burned directly** for heat or transformed into **renewable liquid and gaseous fuels** through various conversion processes.

Sources of Biomass:



- **Wood and wood processing wastes -** This category encompasses firewood, wood pellets, wood chips, and residues from lumber and furniture mills, such as sawdust and black liquor from pulp and paper production.
- **Agricultural crops and waste materials -** This includes crops like corn, soybeans, sugar cane, switchgrass, woody plants, and algae, alongside residues from crop and food processing.
- **Biogenic materials in municipal solid waste -** This comprises products made from paper, cotton, and wool, as well as food, yard, and wood wastes.
- **Animal manure and human sewage -** These organic materials also serve as significant biomass sources.

Conversion Processes for Biomass Energy:

- **Direct Combustion - Direct combustion** is the most prevalent method for converting biomass to energy, where the biomass is burned to generate heat. This technique is commonly employed for heating buildings and water, supplying industrial process heat, and producing electricity through steam turbines.
- **Thermochemical Conversion -** This method involves **thermochemical conversion**, including processes like **pyrolysis** and **gasification**. In these thermal decomposition processes, biomass feedstock materials are heated in **closed, pressurized vessels** called gassifiers, achieving high temperatures that convert the biomass into solid, gaseous, and liquid fuels.
- **Chemical Conversion -** The **chemical conversion** process known as **transesterification** converts vegetable oils, animal fats, and greases into **fatty acid methyl esters (FAME)**, which are utilized to produce **biodiesel**.
- **Biological Conversion - Biological conversion** encompasses methods such as **fermentation** and **anaerobic digestion**. Fermentation transforms biomass into **ethanol**, a viable vehicle fuel, while anaerobic digestion produces **renewable natural gas**, also known as **biogas** or **biomethane**. This process occurs in anaerobic digesters, commonly found at sewage treatment plants and dairy or livestock operations, as well as in landfills where solid waste decomposes.

- **Anaerobic digestion**, or **biomethanation**, involves the breakdown of biodegradable matter by anaerobic microorganisms in an oxygen-free environment, yielding methane-rich biogas and effluent. This process consists of three sequential functions: **hydrolysis**, **acidogenesis**, and **methanogenesis**.

- **Cogeneration - Cogeneration** refers to the simultaneous production of two forms of energy from a single fuel source, typically heat and electricity. In a conventional power plant, fuel is combusted in a boiler to generate high-pressure steam, which drives a turbine for electricity generation. However, conventional plants typically achieve only around **35% efficiency**, as a significant portion of heat is lost during steam condensation.

In contrast, cogeneration plants utilize the **low-pressure exhaust steam** from the turbine for heating purposes, achieving much higher efficiency levels ranging from **75% to 90%**. This approach not only meets the dual demands for power and heat but also offers **cost savings** for the plant and a **reduction in emissions** of pollutants due to decreased fuel consumption.

Advantages of Biomass Energy:

- **Versatility - Biomass energy** can be utilized for various applications, including cooking gas and electricity generation, with minimal environmental impact.
- **Renewable Resource -** Biomass is a **renewable energy source** that is abundantly provided by nature, making it a sustainable option for energy needs.
- **Low Net CO2 Emissions -** The combustion of biomass results in significantly **lower CO2 emissions** compared to conventional fossil fuels, contributing to a reduction in greenhouse gases (GHGs).
- **Reduced Emissions of SO2 and NOx -** Biomass energy generates **fewer sulfur (SO2)** and **nitrogen oxides (NOx)** pollutants than traditional fossil fuels, making it a cleaner alternative.

Disadvantages of Biomass Energy:

1. Low Energy Density/Yield - Certain biofuels, such as **ethanol**, possess a relatively **low energy yield** compared to gasoline. For instance, corn-derived bioethanol may provide little to no net energy unless fortified with fossil fuels.

2. Land Conversion - The production of biomass may require land that could be used for other essential purposes, such as **agriculture, conservation,** or housing, potentially leading to a decline in **food production** and **biodiversity loss**.

3. Environmental Concerns - Intensive agricultural practices associated with biomass production can lead to several environmental issues, including - **Nutrient pollution, Soil depletion, Soil erosion** and **Water pollution**

Meeting India's Energy Demands:

- **Growing Energy Demand -** Bioenergy is pivotal in addressing the **growing energy demands** in India, especially in rural regions. Approximately **25% of India's primary energy** is derived from biomass resources, with nearly **70% of the rural population** relying on biomass for daily energy needs.
- **Climate Change Mitigation -** Biomass energy offers significant advantages over fossil fuels, particularly in terms of **greenhouse gas emissions**. By recycling carbon from the atmosphere, biomass helps in reducing the additional carbon emissions generated from fossil fuel extraction and usage.
- **Market Growth Potential -** The market for **renewable energy systems** in both rural and urban India is projected to grow exponentially. However, bioenergy is often classified as **'non-commercial' energy** and is overlooked in many

energy studies. Plants like **Jatropha, Neem**, and other wild species are identified as potential biodiesel sources in India.

- **Waste to Energy Initiatives -** Biofuels can facilitate **waste-to-wealth** initiatives by utilizing renewable biomass resources such as **municipal solid waste, agricultural residues,** and **forestry by-products**. This potential aids in achieving India's renewable energy goal of **175 GW**.
- **Economic Benefits -** Adopting biofuels can significantly **improve farmers' income** and create **employment opportunities**, contributing positively to the economy.
- **Reducing Energy Imports -** Currently, **46.13% of India's energy demands** are met through imports. Developing bioenergy can reduce this dependency, enhancing **energy security** and **self-reliance** in the country.

Government Initiatives:

- **National Target for Biomass Power -** The **Ministry of New and Renewable Energy (MNRE)** has set an ambitious target to achieve **10 GW of installed biomass power** capacity by 2022.
- **National Policy on Biofuels -** The national policy aims to reach a **20% blending target** of biofuels with fossil-based fuels by **2025**, promoting the use of biofuels in the energy mix.
- **Policies for Cogeneration -** The MNRE has introduced policies for **biomass and bagasse cogeneration** to help meet India's energy needs, offering **financial incentives** and subsidies for biomass projects and sugar mills utilizing this technology.
- **Fiscal Incentives -** The government provides a range of **fiscal incentives**, including -**10-year income tax holidays, Concessional customs and excise duty exemptions** for machinery used in biomass projects and **Sales tax exemptions** in specific states.
- **Waste to Energy Projects -** Various **waste-to-energy projects** are being established to generate energy from **urban, industrial,** and **agricultural waste**, including **slaughterhouse waste** and market discards.
- **National Biomass Repository -** The MNRE plans to establish a **'National Biomass Repository'** through a comprehensive national appraisal program to ensure the availability of biofuels produced from domestic feedstock.
- **National Mission on Biomass Co-firing -** To tackle **air pollution** caused by farm stubble burning and reduce the carbon footprint of thermal power generation, the **Ministry of Power** has proposed a **National Mission** focused on increasing biomass use in coal-based thermal power plants. This mission aligns with the **National Clean Air Programme (NCAP)** and aims to facilitate India's transition towards cleaner energy sources.
    - **Objectives of the Mission:**
        - **Increase co-firing levels** from the current **5%** to higher percentages for more carbon-neutral power generation.
        - Promote **research and development** in boiler design to accommodate increased silica and alkalis from biomass pellets.
        - Address **supply chain constraints** related to biomass pellets and agricultural residues.
        - Consider **regulatory issues** associated with biomass co-firing.

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